# CMOC Evidence Table

## CMOC Evidence Table (typed, span-grounded)

### S001 (2001) — Coordination of Analytic and Similarity-Based Processing Strategies and Expertis

**CMOC** `S001-cmoc-6dc142` · polarity: **mixed** · verifier support: 0.18

> *When dermatological diagnosis learners differ by expertise level, reasoning-process instruction triggers relative reliance on analytic and similarity-based processes, leading to instructionally modifiable reasoning-process reliance.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | dermatological diagnosis domain | Coordination of Analytic and Similarity-Based Processing Strategies and Expertise in Dermatological Diagnosis | resolved |
| Context | learner expertise level | The relative reliance on analytic and similarity-based processes is amenable to instruction and dependent on expertise. | resolved |
| Intervention | reasoning-process instruction | The relative reliance on analytic and similarity-based processes is amenable to instruction and dependent on expertise. | resolved |
| Mechanism_Response | relative reliance on analytic and similarity-based processes | The relative reliance on analytic and similarity-based processes is amenable to instruction and dependent on expertise. | resolved |
| Outcome | instructionally modifiable reasoning-process reliance | The relative reliance on analytic and similarity-based processes is amenable to instruction and dependent on expertise. | resolved |

### S003 (2006) — Giving Learners the Best of Both Worlds: Do Clinical Teachers Need to Guard Agai

**CMOC** `S003-cmoc-a0fd62` · polarity: **positive** · verifier support: 0.78

> *When learners are novice diagnosticians, feature-oriented and similarity-trusting instructions trigger additive clinical reasoning, leading to improved diagnostic performance.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | novice diagnosticians | The results support an additive model of clinical reasoning in which instructions to be feature oriented and to trust similarity improve performance in novice d | resolved |
| Intervention | combined reasoning diagnostic instruction | The results support an additive model of clinical reasoning in which instructions to be feature oriented and to trust similarity improve performance in novice d | resolved |
| Mechanism_Resource | feature-oriented and similarity-trusting instructions | instructions to be feature oriented and to trust similarity improve performance in novice diagnosticians. | resolved |
| Mechanism_Response | additive clinical reasoning | The results support an additive model of clinical reasoning in which instructions to be feature oriented and to trust similarity improve performance | resolved |
| Outcome | improved diagnostic performance | instructions to be feature oriented and to trust similarity improve performance in novice diagnosticians. | resolved |

## S004 (2007) — Teaching from the clinical reasoning literature: combined reasoning strategies h

**CMOC** `S004-cmoc-3b264f` · polarity: **positive** · verifier support: 0.42

> *When novice ECG diagnosticians were given an incorrect diagnostic suggestion intended to induce greater non-analytic reliance, the combined analytic and non-analytic reasoning prompt was interpreted as supporting avoidance of single-mode reasoning, leading to comparatively greater accuracy in the bias-incorrect condition.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | novice ECG diagnosticians | Undergraduate psychology students were recruited from McMaster University to ensure that participants had no previous experience with ECG diagnosis. | resolved |
| Intervention | combined reasoning diagnostic instruction | Half of all participants were instructed to reason in a combined manner and half were given no explicit instruction regarding the diagnostic task. | resolved |
| Intervention | diagnostic suggestion ECG presentation | In Study 2 (n = 48), a third of the test ECGs were presented with a correct diagnostic suggestion, a third with an incorrect suggestion, and a third without a s | resolved |
| Mechanism_Resource | incorrect diagnostic suggestion | Six of the ECGs were presented with no additional information, 6 were presented with the correct diagnosis, and 6 were presented with an incorrect diagnosis. | resolved |
| Mechanism_Resource | combined analytic and non-analytic reasoning prompt | Trust this sense of familiarity, but realise that basing decisions solely on similarity can lead to diagnostic errors. So, don/C213t ‘‘jump the gun’’. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Response | author-interpreted avoidance of single-mode reasoning | there is value for the diagnostician in not getting locked into any single mode of thinking about clinical cases. | resolved |
| Mechanism_Response | intended greater non-analytic reliance | the aim of the diagnostic hypothesis manipulation in Study 2 was to induce people towards greater reliance on their non-analytic tendencies | resolved |
| Outcome | greater accuracy under incorrect suggestion | the key com- parison (combined versus no-instruction within the bias-incorrect condition) revealed a statistically significant difference ( F[1,46] = 4.2, P < 0. | resolved |

## S004 (2007) — Teaching from the clinical reasoning literature: combined reasoning strategies h

**CMOC** `S004-cmoc-7bd054` · polarity: **negative** · verifier support: 0.42

> *When novices who had just learned diagnostic rules received the whatever-strategy-comes-naturally prompt, a counter-indicative feature prompt was intended to trigger greater analytic reliance on features, and limited spontaneous combined reasoning uptake led to lower accuracy in the no-instruction condition.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | novices after diagnostic-rule training | One could easily expect that novices, who have just learned a set of diagnostic rules, might be especially vulnerable to this overly analytic ap- proach. | resolved |
| Intervention | whatever-strategy-comes-naturally prompt | /C212Assign a diagnosis to each ECG using whatever strategy comes naturally to you. Feel free to use the feature list | resolved |
| Intervention | biasing feature ECG test presentation | During the test phase of Study 1, participants were presented with 1 of 2 possible test booklets, each consisting of the same 16 ECGs. | resolved |
| Mechanism_Resource | counter-indicative feature prompt | participants were asked to consider (before trying to diagnose the ECG) whether or not the feature named was present on the recording. | resolved |
| Mechanism_Response | intended greater analytic reliance on features | the aim of the feature manipulation was to induce people towards placing greater reliance on their analytic tendencies | resolved |
| Mechanism_Response | limited spontaneous combined reasoning uptake | students do not spontaneously adopt this approach to clinical reasoning as effectively as they might. | resolved |
| Outcome | lower accuracy with biasing feature under no instruction | accuracy rates achieved when a biasing feature was presented were lower than when no biasing feature was presented in the no-instruction condition ( P < 0.06) | resolved |

## S004 (2007) — Teaching from the clinical reasoning literature: combined reasoning strategies h

**CMOC** `S004-cmoc-b839c8` · polarity: **positive** · verifier support: 0.42

> *When novice ECG diagnosticians faced a counter-indicative feature prompt that biased consideration away from the correct diagnosis, the combined analytic and non-analytic reasoning prompt was interpreted as triggering maintained reasoning balance, leading to no accuracy decline under the biasing feature.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | novice ECG diagnosticians | Undergraduate psychology students were recruited from McMaster University to ensure that participants had no previous experience with ECG diagnosis. | resolved |
| Intervention | combined reasoning diagnostic instruction | Half of all participants were instructed to reason in a combined manner and half were given no explicit instruction regarding the diagnostic task. | resolved |
| Intervention | biasing feature ECG test presentation | During the test phase of Study 1, participants were presented with 1 of 2 possible test booklets, each consisting of the same 16 ECGs. | resolved |
| Mechanism_Resource | counter-indicative feature prompt | participants were asked to consider (before trying to diagnose the ECG) whether or not the feature named was present on the recording. | resolved |
| Mechanism_Resource | combined analytic and non-analytic reasoning prompt | Trust this sense of familiarity, but realise that basing decisions solely on similarity can lead to diagnostic errors. So, don/C213t ‘‘jump the gun’’. | resolved |
| Mechanism_Response | biased consideration away from correct diagnosis | biasing participants /C213consideration of the cases away from the correct diagnosis either through mention of a feature consistent with an incorrect diagnosis | resolved |
| Mechanism_Response | author-interpreted maintained reasoning balance | instruction to adopt a combined reasoning approach can help maintain this balance when faced with chal- lenging cases. | resolved |
| Outcome | no accuracy decline under biasing feature | that the accuracy rates did not differ in the combined reasoning condition ( P > 0.6). | resolved |

## S004 (2007) — Teaching from the clinical reasoning literature: combined reasoning strategies h

**CMOC** `S004-cmoc-c241f2` · polarity: **negative** · verifier support: 0.34

> *When novice ECG diagnosticians received the whatever-strategy-comes-naturally prompt, an incorrect diagnostic suggestion was intended to trigger greater non-analytic reliance; without spontaneous combined reasoning uptake, accuracy was lower with incorrect than correct suggestions in the no-instruction group.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | novice ECG diagnosticians | Undergraduate psychology students were recruited from McMaster University to ensure that participants had no previous experience with ECG diagnosis. | resolved |
| Intervention | diagnostic suggestion ECG presentation | In Study 2 (n = 48), a third of the test ECGs were presented with a correct diagnostic suggestion, a third with an incorrect suggestion, and a third without a s | resolved |
| Intervention | whatever-strategy-comes-naturally prompt | /C212Assign a diagnosis to each ECG using whatever strategy comes naturally to you. Feel free to use the feature list | resolved |
| Mechanism_Resource | incorrect diagnostic suggestion | Six of the ECGs were presented with no additional information, 6 were presented with the correct diagnosis, and 6 were presented with an incorrect diagnosis. | resolved |
| Mechanism_Response | intended greater non-analytic reliance | the aim of the diagnostic hypothesis manipulation in Study 2 was to induce people towards greater reliance on their non-analytic tendencies | resolved |
| Mechanism_Response | limited spontaneous combined reasoning uptake | students do not spontaneously adopt this approach to clinical reasoning as effectively as they might. | resolved |
| Outcome | lower accuracy with incorrect than correct suggestions | there was a significant difference between the bias-incorrect condition and the bias-correct condition in the no-instruction group ( $F[1,46] = 4.3$, $P < 0.05$). | resolved |

## S005 (2007) — The benefits of flexibility: the pedagogical value of instructions to adopt mult

**CMOC** `S005-cmoc-02aa97` · polarity: **positive** · verifier support: 0.78

> *When undergraduate learners have no prior ECG experience, combined reasoning diagnostic instruction provides an instructed combined feature and familiarity strategy, triggering empowered use of multiple diagnostic strategies, leading to improved performance beyond no instructions.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | undergraduate learners with no prior ECG experience | A sample of 48 undergraduate psychology students enrolled at McMaster University participated in this 3-phase experiment for course credit. None of the particip | resolved |
| Intervention | combined reasoning diagnostic instruction | Half the participants in both groups were further instructed to carefully identify all features while trusting guidance provided by feelings of familiarity (a c | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Resource | instructed combined feature and familiarity strategy | carefully identify all features while trusting guidance provided by feelings of familiarity (a combined reasoning strategy). | resolved |
| Mechanism_Response | empowered use of multiple diagnostic strategies | The results emphasise the importance of explicitly empowering students to utilise multiple diagnostic strategies, including non-analytic approaches. | resolved |
| Outcome | immediate and one-week persistence of effects | The effects were observed immediately after learning and following a 1-week delay. | resolved |
| Outcome | improved performance beyond no instructions | instructions to utilise a combined reasoning strategy can improve performance above and beyond what can be expected in the absence of such instructions. | resolved |

## S005 (2007) — The benefits of flexibility: the pedagogical value of instructions to adopt mult

**CMOC** `S005-cmoc-26bf0a` · polarity: **positive** · verifier support: 0.78

> *When the delayed test includes ECGs previously seen during training, one-to-one ECG diagnostic training provides prior exposure to specific ECG exemplars, triggering inferred similarity-based recognition processes, leading to higher accuracy for old than new ECGs.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | delayed test with previously seen and novel ECGs | One week after training, participants were presented with a booklet of 15 ECGs; 8 were novel and 7 had been seen during training | resolved |
| Intervention | one-to-one ECG diagnostic training | all participants were taught general information regarding the 12 leads in an ECG during one-to-one teaching sessions with the experimenter. | resolved |
| Mechanism_Resource | prior exposure to specific ECG exemplars during training | 8 were novel and 7 had been seen during training but were distinct from the ECGs seen during the immediate test phase. | resolved |
| Mechanism_Response | inferred similarity-based recognition processes | The consistency with which /C212old/C213(i.e. previously seen) ECGs were diagnosed more accurately than /C212new/C213ECGs suggests that similarity-based process | resolved |
| Outcome | higher accuracy for old than new ECGs | The higher diagnostic accuracy observed for /C212old/C213ECGs relative to /C212new/C213ECGs, regardless of instructions, suggests that very little exposure is r | resolved |

## S005 (2007) — The benefits of flexibility: the pedagogical value of instructions to adopt mult

**CMOC** `S005-cmoc-7f4d16` · polarity: **positive** · verifier support: 0.84

> *When ECG diagnostic categories are similar and easily confusable, contrastive learning instruction provides Contrastive Diagnostic Scaffolding, triggering possible understanding of symptom diagnosticity and reduced premature closure from feature-overlap awareness, leading to higher diagnostic accuracy than non-contrastive learning.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | confusable ECG diagnostic categories | diagnostic categories that were: 1 similar by virtue of their having features in common, and 2 known to be easily confusable based on the results of previous st | resolved |
| Intervention | contrastive learning instruction | The remaining participants were explicitly instructed to draw comparisons between the diagnostic category being learned and another confusable diagnostic catego | resolved |
| Mechanism_Resource | Contrastive Diagnostic Scaffolding | participants were asked to compare and contrast the features present on each example with those of a normal ECG and with those of a confusable differential. | resolved |
| Mechanism_Response | reduced premature closure from feature-overlap awareness | they were less likely to fall victim to premature closure due to an increased awareness that a feature indicative of a particular disorder can also be indicativ | resolved |
| Mechanism_Response | possibly better symptom diagnosticity understanding | One plausible, although speculative, explanation for this finding is that participants in the contrastive condition better understood the diagnosticity of each | UNRESOLVED |
| Outcome | higher diagnostic accuracy than non-contrastive learning | Participants in the contrastive learning condition performed significantly better than those in the non-contrastive learning condition | UNRESOLVED |
| Outcome | immediate and one-week persistence of effects | The effects were observed immediately after learning and following a 1-week delay. | resolved |

## S005 (2007) — The benefits of flexibility: the pedagogical value of instructions to adopt mult

**CMOC** `S005-cmoc-b59748` · polarity: **negative** · verifier support: 0.78

> *When undergraduate learners have no prior ECG experience and are left unguided, free-choice reasoning prompts trigger self-selected combined reasoning with less benefit, leading to lower accuracy than explicit combined instruction.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | undergraduate learners with no prior ECG experience | A sample of 48 undergraduate psychology students enrolled at McMaster University participated in this 3-phase experiment for course credit. None of the particip | resolved |
| Intervention | Unguided reasoning conditions | The remaining participants were given no instructions on how to approach the diagnostic task. | resolved |
| Mechanism_Resource | free-choice reasoning prompt | Assign a diagnosis to each ECG using whatever strategy comes naturally to you. Feel free to use the feature list, but approach each case using whatever strategy | resolved |
| Mechanism_Response | self-selected combined reasoning with less benefit | Students might naturally utilise a combined reasoning strategy, but they appear to do so with less benefit than when they are provided with concrete instruction | UNRESOLVED |
| Outcome | lower accuracy than explicit combined instruction | Those in the combined reasoning condition significantly outperformed those in the spontaneous reasoning condition | resolved |

## S006 (2008) — Fostering diagnostic knowledge through computer-supported, case-based worked exa

**CMOC** `S006-cmoc-5406b8` · polarity: **positive** · verifier support: 0.86

> *When learners face diagnostic difficulty, computer-supported case-based worked examples provide a diagnostic process schema and worked examples, triggering lower cognitive load and more cognitive resources invested in schema construction, leading to effective and efficient learning.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | diagnostic difficulty among learners | However, students and attending doctors have difficulties ascertaining the right diagnosis. | resolved |
| Intervention | computer-supported case-based worked examples | a case-based, worked example approach was realised in a computer-based learning environment. | resolved |
| Mechanism_Resource | diagnostic process schema and worked examples | The learning environment, integrated in the CASUS learning tool, consisted of a schema illustrating the steps of the diagnostic process and six worked examples | UNRESOLVED |
| Mechanism_Response | more cognitive resources invested in schema construction | Therefore, more cognitive resources can be invested in processes of schema construction. | resolved |
| Mechanism_Response | lower cognitive load | studying worked examples imposes lower levels of cognitive load on the learner than solving training problems, particularly because no extensive search processe | resolved |
| Outcome | effective and efficient worked-example learning | Our results demonstrate that the case-based, worked example approach is effective and efficient. | resolved |

## S006 (2008) — Fostering diagnostic knowledge through computer-supported, case-based worked exa

**CMOC** `S006-cmoc-5db6ab` · polarity: **negative** · verifier support: 0.78

> *When learners with additive, mechanistic diagnostic habits work in complex diagnostic learning environments, erroneous examples with only KCR feedback provide right-or-wrong judgements, triggering deduction from later solution steps, superficial processing and limited deep conceptual understanding, leading to detrimental diagnostic knowledge acquisition.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | additive mechanistic diagnostic approach | medical students at the end of their clinical studies proceed in an additive and mechanistic way; they tend to collect large amounts of data without building co | resolved |
| Context | complex diagnostic learning environments | In fact, at least in the context of complex learning environments, the feedback given must inform students why the respective step in the diagnostic process is | resolved |
| Intervention | erroneous examples with KCR feedback | In the condition /C212errors and KCR feedback/C213the learners had to deduce the right step in the diagnostic process from the subsequent solution step in the e | resolved |
| Mechanism_Resource | severe erroneous examples | In the condition with errors, the student makes several severe errors. | resolved |
| Mechanism_Resource | right-or-wrong only feedback | Considerations, conclusions and further procedures are evaluated only as right or wrong; no further explanation is forthcoming. | resolved |
| Mechanism_Response | deduction from subsequent solution step | the learners had to deduce the right step in the diagnostic process from the subsequent solution step in the example. | resolved |
| Mechanism_Response | limited deep conceptual understanding | For most students, the information that a specific diagnostic conclusion or procedure is not adequate seems not to be sufficient to induce deep conceptual underst | resolved |
| Mechanism_Response | superficial example processing | However, as Renkl 10 showed, a lot of learners process example information rather superficially. | resolved |
| Outcome | detrimental diagnostic knowledge acquisition | When only KCR feedback is given, erroneous examples are detrimental. | resolved |

## S006 (2008) — Fostering diagnostic knowledge through computer-supported, case-based worked exa

**CMOC** `S006-cmoc-8e9884` · polarity: **positive** · verifier support: 0.78

> *When students have insufficiently integrated diagnostic schemata, computer-supported case-based worked examples provide severe erroneous examples plus elaborated GP feedback, triggering effective processing, error identification and correct-solution understanding, leading to improved diagnostic knowledge acquisition.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | insufficiently integrated diagnostic schemata | Furthermore, these components are not sufficiently integrated into more abstract structures (schemata) which would allow effective and efficient diagnostic reason | resolved |
| Intervention | computer-supported case-based worked examples | a case-based, worked example approach was realised in a computer-based learning environment. | resolved |
| Mechanism_Resource | elaborated GP feedback | In the elaborated version of the feedback, the (fictitious) doctor gives additional explanations to the (right or wrong) considerations and further consequences | resolved |
| Mechanism_Resource | severe erroneous examples | In the condition with errors, the student makes several severe errors. | resolved |
| Mechanism_Response | error identification and correct-solution understanding | Therefore, learners must identify the error and understand the correct solution by systematically comparing the incorrect solution with the correct one. | resolved |
| Mechanism_Response | effective processing of additional information | It is clear that effective processing of this additional information is not without cost: in our study, elaborated feedback resulted in a substantial increase i | resolved |
| Outcome | improved diagnostic knowledge acquisition | The results show that the acquisition of diagnostic knowledge is mainly fostered by providing erroneous examples in combination with elaborated feedback. | resolved |

## S006 (2008) — Fostering diagnostic knowledge through computer-supported, case-based worked exa

**CMOC** `S006-cmoc-cfa1c1` · polarity: **positive** · verifier support: 0.78

> *When arterial hypertension is part of the curriculum and assessment, a worked-example learning session provides secondary-hypertension cases and a structured diagnostic schema, triggering cognitive resources for schema construction, leading to a six-week conceptual knowledge effect and MCQ advantage.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | arterial hypertension in curriculum and assessment | Arterial hypertension is part of the curriculum during this semester and is also part of a regular multiple-choice question (MCQ) test. | resolved |
| Intervention | worked-example learning session | After a short introduction, students completed the MCQ test measuring prior knowledge, which was followed by the learning session, in which the students worked | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Resource | secondary hypertension worked examples | six worked examples dealing with pheochromocytoma, renal artery stenosis and primary hyperaldosteronism. | resolved |
| Mechanism_Resource | structured secondary-hypertension diagnostic schema | The additional schema allows the steps of the diagnostic process to be contextualised by providing structured information on all secondary hypertension diseases | resolved |
| Mechanism_Response | more cognitive resources invested in schema construction | Therefore, more cognitive resources can be invested in processes of schema construction. | resolved |
| Outcome | six-week conceptual knowledge effect | Six weeks after the experiment had concluded, medium effects of the learning environment on conceptual knowledge were still significant. | resolved |
| Outcome | arterial-hypertension MCQ advantage | However, on the arterial hypertension questions, experimental group students achieved 5.59 of 7 points (SD = 0.95), whereas control group students achieved only | resolved |

## S007 (2010) — Integrated Cases Section: A Course Designed to Promote Clinical Reasoning in Yea

**CMOC** `S007-cmoc-419908` · polarity: **positive** · verifier support: 0.62

> *When Year 2 medical students undertake the Integrated Cases Section course, a course designed to promote clinical reasoning is intended to activate clinical reasoning, and the course is associated with increased diagnostic reasoning confidence and improved script concordance test performance.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | Year 2 medical students | Integrated Cases Section: A Course Designed to Promote Clinical Reasoning in Year 2 Medical Students | resolved |
| Intervention | Integrated Cases Section course | Integrated Cases Section: A Course Designed to Promote Clinical Reasoning in Year 2 Medical Students | resolved |
| Mechanism_Resource | course designed to promote clinical reasoning | Integrated Cases Section: A Course Designed to Promote Clinical Reasoning in Year 2 Medical Students | resolved |
| Mechanism_Response | intended clinical reasoning activation | Integrated Cases Section: A Course Designed to Promote Clinical Reasoning in Year 2 Medical Students | resolved |
| Outcome | improved script concordance test performance | Students' clinical reasoning showed a significant gain after the Integrated Cases Section on a script concordance test. | resolved |
| Outcome | increased diagnostic reasoning confidence | Students reported increased confidence in their diagnostic reasoning ability during the course and after completion of 2 clerkships. | resolved |

## S008 (2010) — To Think Is Good: Querying an Initial Hypothesis Reduces Diagnostic Error in Med

**CMOC** `S008-cmoc-22e879` · polarity: **positive** · verifier support: 0.28

> *When medical students have an incorrect initial diagnosis, a prompt to query the initial hypothesis triggers thinking about that hypothesis, leading to allowed rectification of the incorrect diagnosis and reduced diagnostic error.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | medical students with incorrect initial diagnosis | These data suggest that querying an initial diagnostic hypothesis does not harm a correct diagnosis but instead allows students to rectify an incorrect diagnosi | resolved |
| Intervention | querying an initial diagnostic hypothesis | These data suggest that querying an initial diagnostic hypothesis does not harm a correct diagnosis but instead allows students to rectify an incorrect diagnosi | resolved |
| Mechanism_Resource | prompt to query initial hypothesis | These data suggest that querying an initial diagnostic hypothesis does not harm a correct diagnosis but instead allows students to rectify an incorrect diagnosi | resolved |
| Mechanism_Response | thinking about initial hypothesis | To Think Is Good: Querying an Initial Hypothesis Reduces Diagnostic Error in Medical Students | resolved |
| Outcome | allowed rectification of incorrect diagnosis | These data suggest that querying an initial diagnostic hypothesis does not harm a correct diagnosis but instead allows students to rectify an incorrect diagnosi | resolved |
| Outcome | reduced diagnostic error in medical students | To Think Is Good: Querying an Initial Hypothesis Reduces Diagnostic Error in Medical Students | resolved |

## S008 (2010) — To Think Is Good: Querying an Initial Hypothesis Reduces Diagnostic Error in Med

**CMOC** `S008-cmoc-4644e2` · polarity: **positive** · verifier support: 0.42

> *When medical students have a correct initial diagnosis, a prompt to query the initial hypothesis triggers thinking about that hypothesis, leading to no harm to the correct diagnosis.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | medical students with correct initial diagnosis | These data suggest that querying an initial diagnostic hypothesis does not harm a correct diagnosis but instead allows students to rectify an incorrect diagnosi | resolved |
| Intervention | querying an initial diagnostic hypothesis | These data suggest that querying an initial diagnostic hypothesis does not harm a correct diagnosis but instead allows students to rectify an incorrect diagnosi | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Resource | prompt to query initial hypothesis | These data suggest that querying an initial diagnostic hypothesis does not harm a correct diagnosis but instead allows students to rectify an incorrect diagnosi | resolved |
| Mechanism_Response | thinking about initial hypothesis | To Think Is Good: Querying an Initial Hypothesis Reduces Diagnostic Error in Medical Students | resolved |
| Outcome | no harm to correct diagnosis | These data suggest that querying an initial diagnostic hypothesis does not harm a correct diagnosis but instead allows students to rectify an incorrect diagnosi | resolved |

## S009 (2011) — A Novel Educational Tool for Teaching Diagnostic Reasoning and Laboratory Data I

**CMOC** `S009-cmoc-275539` · polarity: **positive** · verifier support: 0.86

> *When learners have few initial aspirations to become laboratorians, particular courses or instructors, or learners' discovery of the diagnostic power of laboratory data, can trigger piqued interest in clinical pathology.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | few initial laboratorian aspirations | it is likely that few dreamed of becoming laborator- | |
| ians while growing up. | resolved | | |
| Intervention | particular courses or instructors | interest in clinical pathology is piqued by particular | |
| courses or instructors or when we discover the compelling power | resolved | | |
| Mechanism_Response | discovered diagnostic power of laboratory data | when we discover the compelling power of laboratory data to | |
| uncover and clarify hidden disease. | resolved | | |
| Outcome | piqued interest in clinical pathology | More often, interest in clinical pathology is piqued by particular | |
| courses or instructors or when we discover | resolved | | |

## S009 (2011) — A Novel Educational Tool for Teaching Diagnostic Reasoning and Laboratory Data I

**CMOC** `S009-cmoc-473e24` · polarity: **positive** · verifier support: 0.78

> *When diagnosticians have developed expertise and encounter a cascade of data abnormalities, years of diligent study and practice support logical connection of abnormalities to disease mechanisms, leading to identification of failing organ systems.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | experienced diagnosticians with developed expertise | Experienced diagnosticians, veterinarians, and | |
| physicians, with expertise developed over years of diligent study and practice | resolved | | |
| Context | data abnormality cascade for back-tracing | trace back through a cascade of data abnormalities, | |
| using logic to connect them to underlying mechanisms of disease | resolved | | |
| Intervention | diligent study and practice developing expertise | with expertise developed over years of diligent study and practice, become | |
| increasingly adept at learning to trace back | resolved | | |
| Mechanism_Response | logical connection to disease mechanisms | using logic to connect them to underlying mechanisms of disease | resolved |
| Outcome | identification of failing organ systems | using logic to connect them to underlying mechanisms of disease and ultimately iden- | |
| tifying failing organ systems. | resolved | | |

## S010 (2011) — Bringing Explicit Insight into Cognitive Psychology Features during Clinical Rea

**CMOC** `S010-cmoc-075eb2` · polarity: **positive** · verifier support: 0.42

> *When students are learning clinical reasoning, a case-based clinical reasoning seminar intervention provides a focus on cognitive features of reasoning, intended to trigger insight into those reasoning features, leading to improved aspects of diagnostic competence.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | students learning clinical reasoning | This case-based clinical reasoning seminar intervention, designed to bring students insight into cognitive features of their reasoning, improved aspects of diag | resolved |
| Intervention | case-based clinical reasoning seminar intervention | This case-based clinical reasoning seminar intervention, designed to bring students insight into cognitive features of their reasoning, improved aspects of diag | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Resource | cognitive features of reasoning focus | This case-based clinical reasoning seminar intervention, designed to bring students insight into cognitive features of their reasoning, improved aspects of diag | resolved |
| Mechanism_Response | insight into cognitive features of reasoning | This case-based clinical reasoning seminar intervention, designed to bring students insight into cognitive features of their reasoning, improved aspects of diag | resolved |
| Outcome | improved aspects of diagnostic competence | This case-based clinical reasoning seminar intervention, designed to bring students insight into cognitive features of their reasoning, improved aspects of diag | resolved |

## S011 (2011) — How we created virtual patient cases for primary care-based learning

**CMOC** `S011-cmoc-1f928d` · polarity: **positive** · verifier support: 0.42

> *When students are in a six-week primary care rotation, self-directed VP cases provide autonomous clip control and attempt scores, triggering formative use and progress mapping, leading to repeated use across days and higher end-of-rotation usage.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | six-week primary care rotation setting | During this pilot project, we collected data from three cohorts of students on the 6-week primary care rotation. | resolved |
| Intervention | self-directed VP cases | The VP cases are principally intended for self-directed learning, but could also be used as a teaching tool in a small group or lecture setting. | resolved |
| Mechanism_Resource | attempt scores available for comparison | They can compare their scores on each attempt to map their progress and identify ongoing knowledge/application gaps. | resolved |
| Mechanism_Resource | autonomous repetition or skipping of clips | Participants also have the autonomy to repeat video clips or skip them according to their preferences. | resolved |
| Mechanism_Response | progress mapping and gap identification | They can compare their scores on each attempt to map their progress and identify ongoing knowledge/application gaps. | resolved |
| Mechanism_Response | formative use of VP cases | This suggests that students may be using the VPs in a formative manner. | resolved |
| Outcome | higher end-of-rotation usage | we observed higher student usage towards the end of their primary care rotation, indicating that they are using the VP cases for revision purposes. | resolved |
| Outcome | repeated use across different days | students are multiple users; the average number of attempts at each scenario per student is greater than 1, and consecutive attempts occur on different days. | resolved |

## S011 (2011) — How we created virtual patient cases for primary care-based learning

**CMOC** `S011-cmoc-3014e9` · polarity: **positive** · verifier support: 0.78

> *When short rotations limit longitudinal exposure, time-compressed VP case scenarios provide a compressed longitudinal patient course, triggering observation of test-of-time and test-of-treatment reasoning, leading to theorised construction of clinical reasoning skills.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | short rotations limiting longitudinal exposure | Undergraduate primary care rotations are often short. This means that students are unlikely to observe consultations or meet patients suffering from all of the | resolved |
| Intervention | time-compressed VP case scenarios | VP cases can be used to compress time, allowing students to observe strategies such as ‘ test of time ’o r ‘ test of treatment ’ | resolved |
| Mechanism_Resource | compressed longitudinal patient course | This compression of time also allows the evolution of chronic disease, longitudinal management and treatment effects to be illustrated. | resolved |
| Mechanism_Response | observation of test-of-time reasoning | allowing students to observe strategies such as ‘ test of time ’o r ‘ test of treatment ’ which GPs may use to define their diagnosis | resolved |
| Outcome | theorised construction of clinical reasoning skills | Follow-up of and reflection on patients is a powerful discipline for the learning and construction of clinical reasoning skills | resolved |

## S011 (2011) — How we created virtual patient cases for primary care-based learning

**CMOC** `S011-cmoc-72b9d0` · polarity: **negative** · verifier support: 0.78

> *When VP cases are rolled out to primary care rotation cohorts as optional self-directed access, the self-directed access triggers limited voluntary engagement with cases, leading to restricted uptake among potential students.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | pilot rollout to rotation cohorts | During this pilot project, we collected data from three cohorts of students on the 6-week primary care rotation. | resolved |
| Intervention | optional self-directed VP access | The VP cases are principally intended for self-directed learning, but could also be used as a teaching tool in a small group or lecture setting. | resolved |
| Mechanism_Resource | self-directed access to cases | The VP cases are principally intended for self-directed learning, but could also be used as a teaching tool in a small group or lecture setting. | resolved |
| Mechanism_Response | limited voluntary engagement with cases | Approximately one-third of students of all potential students used the cases, giving a total of 82 students who used the cases. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Outcome | restricted uptake among potential students | Approximately one-third of students of all potential students used the cases, giving a total of 82 students who used the cases. | resolved |

## S011 (2011) — How we created virtual patient cases for primary care-based learning

**CMOC** `S011-cmoc-e1193c` · polarity: **positive** · verifier support: 0.86

> *When GP diagnostic approaches are not explicitly taught, tailored decision-tree VP cases provide virtual experience of these real-time reasoning strategies, triggering practice and refinement of GP reasoning, leading to theorised acquisition of primary care reasoning skills.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | GP diagnostic approaches not explicitly taught | The unique diagnostic approaches used by GPs tend not to be explicitly taught to medical students. | resolved |
| Intervention | tailored decision-tree VP cases | VP cases can be tailored using decision-making software to give students experience of primary care specific clinical reasoning skills. | resolved |
| Mechanism_Resource | virtual experience of GP reasoning strategies | Diagnostic decision tree software can be used to create VP cases that give virtual experience of these real-time clinical reasoning techniques. | resolved |
| Mechanism_Response | practice and refinement of GP reasoning | The VP cases give students opportunities to practise and refine unique clinical reasoning skills typically deployed by general practitioners (GPs) (Balla & Edwa | resolved |
| Outcome | theorised acquisition of primary care reasoning skills | we tailored the cases to promote the acquisition of primary care specific clinical reasoning skills. | resolved |

## S011 (2011) — How we created virtual patient cases for primary care-based learning

**CMOC** `S011-cmoc-f020cb` · polarity: **positive** · verifier support: 0.78

> *When undergraduate primary care rotations are short, video-enriched VP cases provide simulated GP consultations with patient videos, triggering active virtual participation in consultations, leading to theorised enhanced acquisition of primary care clinical reasoning skills.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | short undergraduate primary care rotations | Undergraduate primary care rotations are often short. This means that students are unlikely to observe consultations or meet patients suffering from all of the | resolved |
| Intervention | video-enriched virtual patient cases | The aim of our video enriched VP cases is to simulate GP consultations revolving around these core presentations, giving students the opportunity to be active p | resolved |
| Mechanism_Resource | simulated GP consultations with patient videos | The aim of our video enriched VP cases is to simulate GP consultations revolving around these core presentations, giving students the opportunity to be active p | resolved |
| Mechanism_Response | active virtual participation in consultations | giving students the opportunity to be active participants in the consultations in a virtual way. | resolved |
| Outcome | theorised enhanced primary care reasoning acquisition | with the aim of enhancing acquisition of primary care specific clinical reasoning skills. | resolved |

## S013 (2011) — The influence of medical students selfexplanations on diagnostic performance

**CMOC** `S013-cmoc-172861` · polarity: **positive** · verifier support: 0.78

> *When third-year medical students in clerkship training generated self-explanations without feedback, the intervention provided definitions, examples, and instructions prompting self-explanation, triggering extensive case elaboration and leading to better diagnostic performance after one week.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | third-year medical students in clerkship training | All participants were in Year 3 of the medical undergraduate programme, which comprises 4 years, of which 2.5 years of problem-based learning | resolved |
| Intervention | self-explanations generated without feedback | The self-explanations were generated after minimal instructions and no feedback was provided to students. | resolved |
| Mechanism_Resource | instructions prompting self-explanation | The instructions prompted the participants to self-explain; 10 minutes were sufficient for students to work out the problem in all cases. | UNRESOLVED |
| Mechanism_Resource | definition and example of self-explanation | they were provided with a definition of self-explanation and an example of the procedure. | resolved |
| Mechanism_Resource | clinical audio example of self-explanation | each participant listened to a 7-minute audio example of self-explanation on a clinical case (mono-arthritis) | UNRESOLVED |
| Mechanism_Response | extensive case elaboration | Even without feedback, simply by elaborating more extensively on the cases, students benefited from self-explanation. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Outcome | better diagnostic performance after one week | Generating self-explanations without feedback resulted in better diagnostic performance than in the control group at 1 week after the intervention. | resolved |

## S013 (2011) — The influence of medical students selfexplanations on diagnostic performance

**CMOC** `S013-cmoc-758121` · polarity: **negative** · verifier support: 0.42

> *When clinical cases are more familiar and learners already hold well-defined illness scripts, there is reduced scope for elaboration and reconstruction, so self-explanation diagnostic case solving leads to no diagnostic performance benefit for familiar topics.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | more familiar clinical cases with well-defined scripts | After repeated exposure to a problem, most students will probably hold a well-defined illness script in their memory | UNRESOLVED |
| Intervention | self-explanation diagnostic case solving | having previously engaged in self-explanation in the course of solving problems did not make any difference. | UNRESOLVED |
| Mechanism_Resource | opportunity to generate explanations to oneself | participants were asked to read each case and to generate self-explanations out loud. | resolved |
| Mechanism_Response | reduced scope for elaboration and reconstruction | Less room will be left for the elaboration and reconstruction of knowledge and, therefore, for self-explanation to play a role in learning. | resolved |
| Outcome | no diagnostic performance benefit for familiar topics | having previously engaged in self-explanation in the course of solving problems did not make any difference. | UNRESOLVED |

## S013 (2011) — The influence of medical students selfexplanations on diagnostic performance

**CMOC** `S013-cmoc-ea1bd4` · polarity: **positive** · verifier support: 0.78

> *When students face less familiar clinical cases with less developed illness scripts, the opportunity to generate explanations to oneself triggers active sense-making, symptom linking, and knowledge reconstruction, leading to better delayed diagnostic performance on less familiar cases.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | less familiar clinical cases with less developed scripts | less familiar area, for which their illness scripts may be less developed and therefore more subject to major revision. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Intervention | self-explanation diagnostic case solving | generating self-explanations while solving a clinical case led students to make better diagnoses of problems related to the same clinical topic | resolved |
| Mechanism_Resource | opportunity to generate explanations to oneself | participants were asked to read each case and to generate self-explanations out loud. | resolved |
| Mechanism_Response | active sense-making and symptom linking | students more actively attempt to make sense of the clinical findings in a case and try to construct relationships between signs and symptoms | resolved |
| Mechanism_Response | knowledge elaboration and reconstruction | elaboration on the new information, together with the integration of new information into existing knowledge structures and consequent knowledge reconstruction | resolved |
| Outcome | better delayed diagnostic performance on less familiar cases | demonstrated better diagnostic performance on subsequent clinical cases, but this effect emerged only for cases concerning the less familiar topic. | resolved |

## S014 (2012) — evPaeds: undergraduate clinical reasoning

**CMOC** `S014-cmoc-de591a` · polarity: **positive** · verifier support: 0.78

> *When undergraduate clinical reasoning learners use evPaeds, it provides individual or small-group clinical reasoning practice and is positively received by students, with reported outcomes of student popularity and clinical reasoning practice/development.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | undergraduate clinical reasoning learners | evPaeds: undergraduate clinical reasoning | |

ABSTRACT: evPaeds has proven popular with students to practise and develop clinical reasoning, individually or in sma | resolved | | Intervention | evPaeds clinical reasoning intervention | evPaeds has proven popular with students to practise and develop clinical reasoning, individually or in small group sessions. | resolved | | Mechanism_Resource | individual or small-group clinical reasoning practice | to practise and develop clinical reasoning, individually or in small group sessions. | resolved | | Mechanism_Response | positive student reception of evPaeds | evPaeds has proven popular with students to practise and develop clinical reasoning | resolved | | Outcome | clinical reasoning practice and development | to practise and develop clinical reasoning, individually or in small group sessions. | resolved | | Outcome | popularity among students | evPaeds has proven popular with students to practise and develop clinical reasoning, individually or in small group sessions. | resolved |

## S015 (2012) — Preparing medical students for clinical decision making: A pilot study exploring

**CMOC** `S015-cmoc-0ac6d0` · polarity: **negative** · verifier support: 0.78

> *When students have no prior experience of multitasking situations, multiple tasks and acutely unwell patients trigger stress from pager interruption, leading to abandoned structured patient assessment.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | no prior multitasking situation | The students put these errors down to the fact that they have never been in that situation before. | resolved |
| Intervention | simulated ward environment | As a part of this intervention, students were introduced to a simulated ward environment | resolved |
| Mechanism_Resource | multiple tasks and acutely unwell patients | During their simulation session, the students were faced with multiple tasks and acutely unwell patients, and therefore had to prioritise. | resolved |
| Mechanism_Response | stress from pager interruption | SA3 – It was stressful. When you have a sick patient and then the page goes off, that was stressful. | resolved |
| Outcome | abandoned structured patient assessment | SA3 – When I was going to the patient I was paged, so I forgot to finish the ABCDE. I didn't really assess the patient properly. | resolved |

## S015 (2012) — Preparing medical students for clinical decision making: A pilot study exploring

**CMOC** `S015-cmoc-3b38ac` · polarity: **negative** · verifier support: 0.78

> *When students have little prior prioritisation decision experience, shadowing where the FY1 makes decisions and the student follows triggers feeling cocooned from responsibility, leading to feeling not fully prepared for FY1 decisions.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | little prior prioritisation decision experience | Students had little, if any, experience of making a number of decisions that they will be expected to make when they start work, despite the introduction of a p | resolved |
| Intervention | FY1 shadowing programme | Students had little, if any, experience of making a number of decisions that they will be expected to make when they start work, despite the introduction of a p | resolved |
| Mechanism_Resource | shadowing where FY1 decides and student follows | SA2 – On Preparation for Practice I have my FY1 and she makes all the decisions and I just follow her around | resolved |
| Mechanism_Response | feeling cocooned from responsibility | SA2 – When you are a medical student you are kind of cocooned, wrapped up in cotton wool | resolved |
| Outcome | feeling not fully prepared for FY1 decisions | Students felt that shadowing FY1s had not fully prepared them for making decisions as FY1s. | resolved |

## S015 (2012) — Preparing medical students for clinical decision making: A pilot study exploring

**CMOC** `S015-cmoc-7bb0b0` · polarity: **positive** · verifier support: 0.62

> *When students are familiar with existing structured tools, a visible SBAR referral prompt triggers reduced fear and more comfortable senior communication, leading to greater SBAR use.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | familiarity with existing structured tools | Students found that existing tools, with which they were familiar, were very useful and helped to reduce their stress levels. | resolved |
| Intervention | simulated referral task with SBAR support | During the simulator session, the SBAR tool was written down next to the phones. Making it readily available increased the use of SBAR. | resolved |
| Mechanism_Resource | visible structured SBAR referral prompt | During the simulator session, the SBAR tool was written down next to the phones. Making it readily available increased the use of SBAR. | resolved |
| Mechanism_Response | more comfortable structured senior communication | The students felt more comfortable using the structured, SBAR approach when it came to actually speaking to their senior on the phone. | resolved |
| Mechanism_Response | reduced fear during referral | SB2 – The SBAR structure was really useful. It took a bit of the fear away definitely | resolved |
| Outcome | greater SBAR use | During the simulator session, the SBAR tool was written down next to the phones. Making it readily available increased the use of SBAR. | resolved |

## S015 (2012) — Preparing medical students for clinical decision making: A pilot study exploring

**CMOC** `S015-cmoc-8c0491` · polarity: **positive** · verifier support: 0.78

> *When advanced students nearing FY1 transition feel unprepared for clinical decision making, the tool, tutorial and simulated ward scenarios trigger positive adaptation and learning, leading to increased subjective preparedness for FY1 decision making.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | students feeling unprepared for FY1 decision making | Despite extended shadowing programmes, students feel unprepared for clinical decision making as FY1s, and lack effective decision making strategies. | resolved |
| Context | advanced students nearing FY1 transition | The prospective participants had all completed their final university examinations, and this was their last undergraduate placement before they graduated and st | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Intervention | tripartite clinical decision making teaching | A teaching intervention was developed, consisting of a clinical decision making tool, a tutorial and scenarios within a simulated ward environment. | resolved |
| Mechanism_Resource | tool tutorial and simulated ward scenarios | A teaching intervention was developed, consisting of a clinical decision making tool, a tutorial and scenarios within a simulated ward environment. | resolved |
| Mechanism_Response | positive adaptation and learning | The students were overwhelmingly positive about the individual simulation session, and showed evidence of quickly adapting and learning how to prioritise and ma | resolved |
| Outcome | increased subjective preparedness for FY1 decision making | Experiencing complex decision making scenarios through individually orientated simulation results in students being subjectively more prepared for work as FY1s. | resolved |

## S015 (2012) — Preparing medical students for clinical decision making: A pilot study exploring

**CMOC** `S015-cmoc-ac6fbf` · polarity: **negative** · verifier support: 0.78

> *When learners lack familiarity with AM I HERO NOW, the memorable mnemonic decision prompt triggers perceived time and complexity burden, leading to the tool not being used in simulated practice.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | lack of familiarity with AM I HERO NOW | SB5 – I had never used it before. When I went in, the aide memoire went out the window. | resolved |
| Intervention | AM I HERO NOW decision tool instruction | a decision making tool (AM I HERO NOW) was introduced to students, to be used when faced with making a decision | resolved |
| Mechanism_Resource | memorable mnemonic decision prompt | The tool was designed to be a memorable mnemonic, incorporating multiple aspects of clinical decision making, with the aim of prompting students to avoid certai | resolved |
| Mechanism_Response | perceived time and complexity burden | There were three main reasons why the students did not use it; they felt it would take too long to use, it was too complex and they were unfamiliar with it. | resolved |
| Outcome | AM I HERO NOW not used in simulated practice | However, the ‘AM I HERO NOW’ tool, although introduced, was not actually used by students as a decision making tool during their simulated practice in this stud | resolved |

## S016 (2012) — Reflection as a strategy to foster medical students acquisition of diagnostic co

**CMOC** `S016-cmoc-12f9af` · polarity: **negative** · verifier support: 0.62

> *When learners have underdeveloped illness scripts, demanding step-by-step case comparison may trigger high cognitive load, exhaustion, or confusion, leading to poorer immediate diagnostic performance.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | learners with underdeveloped illness scripts | Therefore, they are not yet likely to have well-developed illness scripts. | resolved |
| Intervention | Structured case-based reasoning | This article reports a study on the effects of using structured reflection as an instructional strategy on medical students' learning of clinical diagnosis. | resolved |
| Mechanism_Resource | demanding step-by-step case comparison | students in the structured reflection condition went through the step-by-step process of matching findings from the case to each diagnosis and identifying contrad | resolved |
| Mechanism_Response | possible high cognitive load, exhaustion, or confusion | reflection imposes a high cognitive load during the learning phase and leads to exhaustion, or that reflection leads to initial confusion | resolved |
| Outcome | poorer immediate diagnostic performance | On the test that immediately followed the learning phase, however, students in the reflection condition performed worse than students who generated an immediate | resolved |

## S016 (2012) — Reflection as a strategy to foster medical students acquisition of diagnostic co

**CMOC** `S016-cmoc-b4316f` · polarity: **negative** · verifier support: 0.62

> *When learners have underdeveloped illness scripts, speeded single-diagnosis instruction lowers the likelihood of reflective processing, leading to no significant signs of immediate-test learning in the non-reflection conditions.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | learners with underdeveloped illness scripts | Therefore, they are not yet likely to have well-developed illness scripts. | resolved |
| Intervention | immediate diagnosis case practice | In the immediate diagnosis condition, participants were requested to read the case and write down the most likely diagnosis for the case. | resolved |
| Mechanism_Resource | speeded single-diagnosis instruction | Both accuracy and speed were stressed, and participants were instructed to move on to the next page to solve a word puzzle after they had made a diagnosis. | resolved |
| Mechanism_Response | lower likelihood of engaging in reflection | This procedure was used to minimise the likelihood that participants would engage in reflection in the other experimental conditions. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Outcome | no significant immediate-test learning signs in non-reflection conditions | students in the other conditions showed no significant signs of learning on the immediate test. | resolved |

## S016 (2012) — Reflection as a strategy to foster medical students acquisition of diagnostic co

**CMOC** `S016-cmoc-de8805` · polarity: **positive** · verifier support: 0.78

> *When Year 4 students have theoretical knowledge but limited patient exposure, structured contrastive and absent-feature reflection prompts trigger elaboration and knowledge reorganisation, leading to improved illness-script quality and better delayed diagnostic transfer.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | Year 4 students with theoretical knowledge but limited patient exposure | We recruited Year 4 students because students at this point in their training have been exposed to theoretical knowledge about diseases but not yet to many pati | resolved |
| Intervention | Structured case-based reasoning | This article reports a study on the effects of using structured reflection as an instructional strategy on medical students' learning of clinical diagnosis. | resolved |
| Mechanism_Resource | absent-feature reflection prompt | (iii) list the findings expected to be present if the diagnosis were correct, but which were not described in the case. | resolved |
| Mechanism_Resource | alternative-diagnosis reflection prompts | They were subsequently asked to list alternative diagnoses, assuming that their initial diagnosis would prove to be incorrect | resolved |
| Mechanism_Resource | Contrastive Diagnostic Scaffolding | They were requested to: (i) list the findings in the case description that support this diagnosis; (ii) list the findings that speak against this diagnosis | resolved |
| Mechanism_Response | elaboration and knowledge reorganisation | This process of elaboration on the case information and search for meaning helps to activate additional relevant knowledge and fosters the integration and reorg | resolved |
| Outcome | improved illness-script quality | However, as a result of elaborate processing, the quality of illness scripts is improved, which leads to better longer-term performance. | resolved |
| Outcome | improved delayed diagnostic transfer | After 1 week, students who had been instructed to reflect during the learning phase outperformed students in the other conditions on new exemplars of the same di | resolved |

## S016 (2012) — Reflection as a strategy to foster medical students acquisition of diagnostic co

**CMOC** `S016-cmoc-ed5d32` · polarity: **negative** · verifier support: 0.62

> *When learners have underdeveloped illness scripts, an alternative-diagnosis generation prompt likely triggers only partial reflective processing, leading to less delayed learning than structured reflection and a significant immediate-to-delayed decrease within that condition.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | learners with underdeveloped illness scripts | Therefore, they are not yet likely to have well-developed illness scripts. | resolved |
| Intervention | differential diagnosis case practice | In the differential diagnosis condition, participants were requested to read the case and write down the most likely diagnosis for the case. | resolved |
| Mechanism_Resource | alternative-diagnosis generation prompt | they were asked to think about alternative diagnoses if their initial hypothesis were to prove incorrect, write them down and, then, draw a conclusion | resolved |
| Mechanism_Response | likely partial reflective processing | Generating alternative diagnoses is likely to have led the students in the differential diagnosis group to also engage in reflection to some degree. | resolved |
| Outcome | significant immediate-to-delayed decrease within differential condition | decreased significantly in the differential diagnosis condition ( t(15 = 2.54, p = 0.012), and increased significantly in the reflection condition | resolved |
| Outcome | less delayed learning than structured reflection | students in the differential diagnosis condition did not learn as much as those in the reflection condition. | resolved |

## S017 (2012) — Should we teach using schemas? Evidence from a randomised trial

**CMOC** `S017-cmoc-0af9d0` · polarity: **negative** · verifier support: 0.78

> *When students face an unknown systolic murmur diagnostic task, diagnosis-specific feature tables can trigger incorrect feature identification that derails reasoning, leading to lower diagnostic accuracy relative to schema-based instruction.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | unknown systolic murmur diagnostic task | However, when students face a patient with an unknown systolic murmur, they must integrate all of this information to arrive at a diagnosis | resolved |
| Intervention | traditional cardiac physical diagnosis teaching | traditional teaching of cardiac physical diagnosis involves diagnosis-specific lists of features, structured around steps in the cardiac physical examination. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Resource | diagnosis-specific feature table | The traditional framework (Fig. 1) was structured around individual valvular lesions and listed the 10 clinical features of each diagnosis in a table format. | resolved |
| Mechanism_Response | incorrect feature identification derails reasoning | Firstly, the traditional instruction group identified more incorrect features. Incorrect features may have a derailing effect on diagnostic reasoning. | resolved |
| Outcome | lower diagnostic accuracy relative to schema | Schema-based instruction was associated with greater diagnostic accuracy (61% versus 26%; mean difference = 35%, 95% confidence interval [CI] 22–47; t 205 = 5.40 | resolved |

## S017 (2012) — Should we teach using schemas? Evidence from a randomised trial

**CMOC** `S017-cmoc-0d224c` · polarity: **positive** · verifier support: 0.78

> *When cardiac auscultation novice students receive schema-based instruction, feature-based diagnostic support triggers uptake of schema features during reasoning, leading to higher diagnostic success for taught lesions.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | cardiac auscultation novice students | Our results provide strong support for the use of schema-based instruction in cardiac auscultation among novice students. | resolved |
| Intervention | schema-based instruction | We aimed to test the effects of schema-based instruction in comparison with those of a traditional framework on the acquisition of cardiac auscultation skills a | resolved |
| Mechanism_Resource | feature-based diagnostic support | The schema framework (Fig. 2) grouped diagnoses according to clinical features based on murmur location and timing | resolved |
| Mechanism_Response | Structured reasoning uptake | schema-instructed students were more likely to report features from the schema in arriving at their diagnosis. | resolved |
| Outcome | higher taught-lesion diagnostic success | Diagnostic success was higher in the schema-based instruction group for taught lesions (mean difference = 38%, 95% confidence interval [CI] 20–56; p < 0.001) | resolved |

## S017 (2012) — Should we teach using schemas? Evidence from a randomised trial

**CMOC** `S017-cmoc-1c92e6` · polarity: **positive** · verifier support: 0.78

> *When cardiac auscultation novice students receive schema-based distinguishing features, the resource triggers reduced incorrect feature identification, leading to higher diagnostic success for untaught diagnostic lesions.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | cardiac auscultation novice students | Our results provide strong support for the use of schema-based instruction in cardiac auscultation among novice students. | resolved |
| Intervention | schema-based instruction | Year 2 medical students were randomised to learn four cardiac diagnoses using schema-based instruction ( n = 26) or traditional instruction ( n = 27) | resolved |
| Mechanism_Resource | schema-based distinguishing features | Schema-based features may be more helpful in the diagnostic process than non-schema-based features as they help distinguish diagnoses. | resolved |
| Mechanism_Response | reduced incorrect feature identification | For untaught diagnoses, schema-instructed students identified fewer overall features, fewer correct features not in the schema and fewer incorrect features (Tabl | resolved |
| Outcome | higher success for untaught diagnostic lesions | Diagnostic success was higher in the schema-based instruction group for taught lesions (mean difference = 38%, 95% confidence interval [CI] 20–56; p < 0.001) and | resolved |

## S017 (2012) — Should we teach using schemas? Evidence from a randomised trial

**CMOC** `S017-cmoc-32739a` · polarity: **negative** · verifier support: 0.86

> *When novice learners receive schema grouping by clinical variables, the resource triggers structured rather than factual knowledge change, leading to no factual knowledge score advantage.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | novice learners | Thirdly, only novice learners participated. It is unclear how our findings might apply to experienced learners, who may already possess compiled knowledge. | resolved |
| Intervention | schema-based instruction | We aimed to test the effects of schema-based instruction in comparison with those of a traditional framework on the acquisition of cardiac auscultation skills a | resolved |
| Mechanism_Resource | schema grouping by clinical variables | The schema framework (Fig. 2) grouped diagnoses according to clinical features based on murmur location and timing | resolved |
| Mechanism_Response | structured-not-factual knowledge change | schemas modified knowledge organisation by increasing the amount of structured knowledge without changing factual knowledge of diagnoses. | resolved |
| Outcome | no factual knowledge score advantage | Schema-based instruction was associated with higher scores on structured knowledge questions (74% versus 55%; p < 0.0001), but not on factual knowledge question | resolved |

## S017 (2012) — Should we teach using schemas? Evidence from a randomised trial

**CMOC** `S017-cmoc-38023d` · polarity: **positive** · verifier support: 0.78

> *When pre-clerkship novice learners receive schema-based instruction, linked and structured clinical information triggers modified structured knowledge organisation, leading to higher structured knowledge scores.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | pre-clerkship novice learners | To our knowledge, this study is the first prospective, randomised study to evaluate schema-based instruction among pre-clerkship learners. | resolved |
| Intervention | schema-based instruction | Year 2 medical students were randomised to learn four cardiac diagnoses using schema-based instruction ( n = 26) or traditional instruction ( n = 27) | resolved |
| Mechanism_Resource | structured clinically relevant information | schemas present the information in a structured, clinically relevant way. | resolved |
| Mechanism_Resource | schemas linking interacting elements | Schemas reduce cognitive load by presenting the most important links between interacting elements. | resolved |
| Mechanism_Response | modified structured knowledge organisation | schemas modified knowledge organisation by increasing the amount of structured knowledge without changing factual knowledge of diagnoses. | resolved |
| Outcome | higher structured knowledge scores | Schema-based instruction was associated with higher scores on structured knowledge questions (74% versus 55%; p < 0.0001), but not on factual knowledge question | resolved |

## S018 (2012) — What and how advanced medical students learn

**CMOC** `S018-cmoc-153de8` · polarity: **negative** · verifier support: 0.78

> *When advanced medical students near internships are given an unguided multiple-case learning task, the open diagnose-explain-and-learn instruction triggers limited overt demonstration of learning in some students, leading to doubts about their ability to exploit case learning fully.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | advanced medical students near internships | Data were collected at the end of the academic year, which means that the students had almost completed the preclinical part of the curriculum and were about to | resolved |
| Intervention | unguided multiple-case learning task | No suggestions were given as to how the students could approach their learning tasks. | resolved |
| Mechanism_Resource | open diagnose-explain-and-learn instruction | their task was to diagnose and explain each case and to learn as much as they could from all these cases. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Response | limited overt demonstration of learning | The other 16 students may have learned but did not show this in their approach to the case, or in their learning reports. | resolved |
| Outcome | limited ability to exploit case learning | As many students did not show any explicit learning processes, this result raises doubts about the students' ability to take full advantage of learning from cas | resolved |

## S018 (2012) — What and how advanced medical students learn

**CMOC** `S018-cmoc-4a3129` · polarity: **positive** · verifier support: 0.86

> *When students have prior endocrine training, a similar-case series provides variation in presentations and causes, triggering comparison of prior knowledge with case findings and other cases, leading to improved knowledge of enabling conditions and consequences.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | students with prior endocrine training | All students repeatedly studied the basic mechanisms and clinical consequences of the endocrinological problems in prior training. | resolved |
| Intervention | similar-case series for diagnosis and explanation | The students were told that they would be presented with a series of cases and that their task was to diagnose and explain each case | resolved |
| Mechanism_Resource | case variation in presentations and causes | We developed two series of five cases with the same underlying pathophysiological process but with different presentations and causes. | resolved |
| Mechanism_Response | comparison of prior knowledge and cases | Some students compared prior knowledge with case findings, some actively compared cases, drawing conclusions about enabling conditions and consequences, whilst o | resolved |
| Outcome | improved enabling-condition and consequence knowledge | It was found that much learning was implicit, whilst students explicitly paid attention to variations in enabling conditions and consequences, which led to impr | resolved |

## S018 (2012) — What and how advanced medical students learn

**CMOC** `S018-cmoc-d695ff` · polarity: **positive** · verifier support: 0.86

> *When advanced medical students are near internships, reduced-feedback similar-case reasoning provides multiple cases with the same underlying problem and official diagnosis feedback, triggering faster and more secure case representation, which is associated with implicit learning and strengthened concept links.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | advanced medical students near internships | Data were collected at the end of the academic year, which means that the students had almost completed the preclinical part of the curriculum and were about to | resolved |
| Intervention | reduced-feedback similar-case reasoning | we asked medical students who are about to enter their internships to reason through a set of cases with the same underlying problem | resolved |
| Mechanism_Resource | official diagnosis feedback after each case | After the students had completed a case, the official diagnosis was given, as they would have received similar feedback if they had encountered the patient as cl | resolved |
| Mechanism_Resource | multiple cases with same underlying problem | We developed two series of five cases with the same underlying pathophysiological process but with different presentations and causes. | resolved |
| Mechanism_Response | faster and more secure case representation | They built up representations faster, became more secure, used more specific language, and tested their knowledge against case information. | UNRESOLVED |
| Outcome | implicit learning and strengthened concept links | we may conclude that there is much implicit learning, which may lead to faster recognition, strengthening of the links between concepts, and tuning of knowledge | resolved |

## S018 (2012) — What and how advanced medical students learn

**CMOC** `S018-cmoc-da8c21` · polarity: **mixed** · verifier support: 0.78

> *When students have incomplete and error-prone knowledge, a similar-case series provides multiple related endocrine cases that can trigger active biomedical debugging effort, leading in some students to partial remediation of misconceptions while others ignore errors.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | incomplete and error-prone knowledge | The other 24 had basic relevant knowledge that was, however, incomplete and error-prone. | resolved |
| Intervention | similar-case series for diagnosis and explanation | The students were told that they would be presented with a series of cases and that their task was to diagnose and explain each case | resolved |
| Mechanism_Resource | multiple related endocrine cases | We developed two series of five cases with the same underlying pathophysiological process but with different presentations and causes. | resolved |
| Mechanism_Response | active biomedical debugging effort | Fourteen engaged in active biomedical reasoning aimed at improving errors or uncertainties the students had recognised. | resolved |
| Outcome | partial remediation of misconceptions | Six participants appeared able to—partially—remediate the misconceptions they had expressed, whereas 2 participants chose to further ignore the errors they noti | resolved |

## S018 (2012) — What and how advanced medical students learn

**CMOC** `S018-cmoc-e84563` · polarity: **negative** · verifier support: 0.78

> *When students have very poor, incomplete, or error-prone prior knowledge, unsupported similar-case reasoning with official diagnosis feedback only is insufficient for debugging unresolved reasoning errors, leading to hampered biomedical fault integration.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | very poor prior knowledge | No participant had all of the relevant knowledge available; 7 had very poor prior knowledge. | resolved |
| Context | incomplete and error-prone knowledge | The other 24 had basic relevant knowledge that was, however, incomplete and error-prone. | resolved |
| Intervention | unsupported similar-case reasoning with reduced feedback | reasoning through a series of cases that have a similar underlying biomedical problem and where only reduced feedback is provided. | resolved |
| Mechanism_Resource | official diagnosis feedback only | After the students had completed a case, the official diagnosis was given, as they would have received similar feedback if they had encountered the patient as cl | resolved |
| Mechanism_Response | unresolved reasoning errors | The remaining 14 participants were either not aware that they made errors of reasoning or were incapable of solving them | resolved |
| Outcome | hampered biomedical fault integration | Integration of biomedical knowledge into the fault component was, however, hampered by lack of knowledge and by misconceptions that students were unable to debu | resolved |

## S019 (2013) — Effect of enhanced analytic reasoning on diagnostic accuracy: A randomized contr

**CMOC** `S019-cmoc-776181` · polarity: **positive** · verifier support: 0.86

> *When advanced medical students near licensure diagnose OSCE standardized-patient cases, a structured analytic-reasoning table and differential-diagnosis prompt trigger consideration of alternatives and conflicting findings, leading to higher diagnostic accuracy.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | advanced medical students near licensure | this study was performed among fourth-year medical students just before attaining the license of medical doctor. | resolved |
| Intervention | enhanced analytic reasoning OSCE instruction | we evaluated the value of enhancing analytic reasoning compared with routine, usual reasoning in medical students. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Resource | differential-diagnosis compatibility and refutation prompt | Students in the analytic reasoning group were asked to complete the table with differential diagnoses and symptoms or signs compatible with or differing from ea | resolved |
| Mechanism_Resource | Structured reasoning support table | After encountering patients, students in the analytic reasoning group received an answer sheet containing a table designed to enhance analytic reasoning. | resolved |
| Mechanism_Response | consideration of alternatives and conflicting findings | Completing the table presented on the answer sheet forced the consideration of alternative diagnoses and of findings that conflicted with a suspected diagnosis. | resolved |
| Outcome | higher diagnostic accuracy | Mean diagnostic accuracy scores were significantly higher in the analytic reasoning group than in the control group (3.40 /C6 0.66 versus 3.05 /C6 0.98; 95% CI | resolved |

## S019 (2013) — Effect of enhanced analytic reasoning on diagnostic accuracy: A randomized contr

**CMOC** `S019-cmoc-a18f8d` · polarity: **negative** · verifier support: 0.78

> *When a clinical presentation is typical, students' heuristic pattern recognition may be sufficient, so the structured analytic-reasoning table can still trigger alternative consideration but adds no diagnostic-accuracy benefit for that case.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | typical acute pyelonephritis presentation | Because the clinical presentation of this case was typical, students who took a heuristic ( pattern recognition) approach may not have made diagnostic errors. | resolved |
| Intervention | enhanced analytic reasoning OSCE instruction | we evaluated the value of enhancing analytic reasoning compared with routine, usual reasoning in medical students. | resolved |
| Mechanism_Resource | Structured reasoning support table | After encountering patients, students in the analytic reasoning group received an answer sheet containing a table designed to enhance analytic reasoning. | resolved |
| Mechanism_Response | sufficient heuristic pattern recognition | Because the clinical presentation of this case was typical, students who took a heuristic ( pattern recognition) approach may not have made diagnostic errors. | resolved |
| Mechanism_Response | consideration of alternatives and conflicting findings | Completing the table presented on the answer sheet forced the consideration of alternative diagnoses and of findings that conflicted with a suspected diagnosis. | resolved |
| Outcome | no added benefit for acute pyelonephritis case | In this study, diagnostic accuracy regarding the case of acute pyelonephritis was similar between the groups. | resolved |

## S019 (2013) — Effect of enhanced analytic reasoning on diagnostic accuracy: A randomized contr

**CMOC** `S019-cmoc-b1cfa7` · polarity: **negative** · verifier support: 0.78

> *When advanced medical students near licensure are in a control condition with an answer sheet lacking the analytic table, the request for one most probable diagnosis is associated with diagnosis without analytic listing processes, leading to lower accuracy relative to the analytic-reasoning group.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | answer sheet lacking analytic table | Students in the control group were simply asked to provide one most probable diagnosis on an answer sheet that lacked the table provided to the test group. | resolved |
| Context | advanced medical students near licensure | this study was performed among fourth-year medical students just before attaining the license of medical doctor. | resolved |
| Intervention | Unguided reasoning conditions | Instead, we evaluated the value of enhancing analytic reasoning compared with routine, usual reasoning in medical students. | resolved |
| Mechanism_Resource | request for one most probable diagnosis | Students in the control group were simply asked to provide one most probable diagnosis on an answer sheet that lacked the table provided to the test group. | resolved |
| Mechanism_Response | diagnosis without analytic listing processes | The analytic reasoning group was requested to list differential diagnoses and findings compatible or not compatible with each diagnosis prior to providing a dia | resolved |
| Outcome | lower diagnostic accuracy relative to analytic reasoning | Mean diagnostic accuracy scores were significantly higher in the analytic reasoning group than in the control group (3.40 /C6 0.66 versus 3.05 /C6 0.98; 95% CI | resolved |

## S019 (2013) — Effect of enhanced analytic reasoning on diagnostic accuracy: A randomized contr

**CMOC** `S019-cmoc-c397a4` · polarity: **negative** · verifier support: 0.42

> *When a case is too complex relative to students' medical knowledge, analytic-reasoning prompts may trigger analytic reasoning, but the knowledge–complexity mismatch constrains benefit, leading to analytic reasoning being ineffective.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | case complexity exceeds students' medical knowledge | analytic reasoning may not be helpful if a case is too complex with respect to the level of students' medical knowledge. | resolved |
| Intervention | analytic reasoning exercised by medical students | Another study has demonstrated contradictory results, suggesting that analytic reasoning exercised by medical students was ineffective in complex cases, but eff | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Resource | differential-diagnosis compatibility and refutation prompt | Students in the analytic reasoning group were asked to complete the table with differential diagnoses and symptoms or signs compatible with or differing from ea | resolved |
| Mechanism_Response | analytic reasoning exercised in complex cases | suggesting that analytic reasoning exercised by medical students was ineffective in complex cases, but effective in simple cases (Mamede et al. 2010a). | resolved |
| Outcome | analytic reasoning not helpful for overly complex cases | That study's findings suggest that analytic reasoning may not be helpful if a case is too complex with respect to the level of students' medical knowledge. | resolved |

## S020 (2013) — Students selfexplanations while solving unfamiliar cases: the role of biomedical

**CMOC** `S020-cmoc-1b80d7` · polarity: **negative** · verifier support: 0.78

> *When students' specific knowledge is less elaborate and they work from clinical case summaries, self-explanation diagnostic case solving provides an opportunity to generate explanations to oneself, triggering active but error-prone knowledge construction and staying closer to the case text, leading to more paraphrases and errors.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | clinical case summaries | Case summaries, approximately 300 words, included the chief complaint, patient's background, medical history, physical examination and the results of laboratory | resolved |
| Context | less elaborate specific knowledge | We assume that in the learning phase, when specific knowledge is less elaborate, there may be more room for self-explanation to play a role in learning. | resolved |
| Intervention | self-explanation diagnostic case solving | They were then asked to read each case while self-explaining aloud for 8 minutes and then they had 2 minutes to write down their diagnosis, together with two su | resolved |
| Mechanism_Resource | opportunity to generate explanations to oneself | Self-explanation is a constructive learning strategy that consists of generating explanations to oneself in response to learning material. | resolved |
| Mechanism_Response | staying closer to case text | students' specific knowledge is less elaborate so that they initially tend to stay closer to the text while self-explaining. | resolved |
| Mechanism_Response | active but error-prone knowledge construction | The greater number of errors in less familiar cases, while few in absolute numbers, probably reflects the fact that specific knowledge is still being actively con | resolved |
| Outcome | more paraphrases and errors | They provided significantly more paraphrases (p = 0.001) and made more errors (p = 0.008). | resolved |

## S020 (2013) — Students selfexplanations while solving unfamiliar cases: the role of biomedical

**CMOC** `S020-cmoc-4ed174` · polarity: **positive** · verifier support: 0.62

> *When third-year medical students work with less familiar jaundice cases, self-explanation diagnostic case solving provides opportunities to generate explanations to oneself, triggering more extensive active thinking and greater self-explanation production, which the authors link to a positive effect on unfamiliar-case diagnosis.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | third-year medical students | Thirty-six third-year medical students from the University of Sherbrooke, Sherbrooke, Quebec, Canada participated in the initial study. | resolved |
| Context | less familiar jaundice cases | Heart failure was identified as a familiar topic and jaundice as a less familiar topic. | resolved |
| Intervention | self-explanation diagnostic case solving | They were then asked to read each case while self-explaining aloud for 8 minutes and then they had 2 minutes to write down their diagnosis, together with two su | resolved |
| Mechanism_Resource | opportunity to generate explanations to oneself | Self-explanation is a constructive learning strategy that consists of generating explanations to oneself in response to learning material. | resolved |
| Mechanism_Response | greater self-explanation production | Students produced more self-explanation segments on less familiar cases than on familiar ones. | resolved |
| Mechanism_Response | more extensive active thinking | Lack of familiarity with a case seems to stimulate medical students to engage in more extensive thinking during self-explanation. | resolved |
| Outcome | positive effect on unfamiliar-case diagnosis | This may clarify the previously found positive effect that self-explanation has on the diagnosis of unfamiliar cases. | resolved |

## S020 (2013) — Students selfexplanations while solving unfamiliar cases: the role of biomedical

**CMOC** `S020-cmoc-d5fb36` · polarity: **positive** · verifier support: 0.78

> *When clinical cases are less familiar and sufficiently challenging, self-explanation diagnostic case solving provides opportunities to generate explanations to oneself, triggering biomedical inference generation and linking of biomedical with clinical knowledge, which may lead to a coherent disease representation and easier future diagnostic retrieval.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | challenging clinical cases | Cases chosen should be challenging enough to force students to revisit underlying basic biomedical knowledge. | resolved |
| Context | less familiar jaundice cases | Heart failure was identified as a familiar topic and jaundice as a less familiar topic. | resolved |
| Intervention | self-explanation diagnostic case solving | They were then asked to read each case while self-explaining aloud for 8 minutes and then they had 2 minutes to write down their diagnosis, together with two su | resolved |
| Mechanism_Resource | opportunity to generate explanations to oneself | Self-explanation is a constructive learning strategy that consists of generating explanations to oneself in response to learning material. | resolved |
| Mechanism_Response | biomedical inference generation | When solving less familiar cases, students generated significantly more biomedical inferences than with familiar cases. | resolved |
| Mechanism_Response | linking biomedical and clinical knowledge | with less familiar cases, self-explanation helps students apply biomedical knowledge and create new links with clinical knowledge. | resolved |
| Outcome | coherent disease representation | self-explanation as a learning strategy for clinical reasoning works better with less familiar cases because it provides opportunities to link biomedical to cli | resolved |
| Outcome | easier future diagnostic retrieval | The resulting more coherent mental representation of disease may then be retrieved more easily during future diagnostic problem-solving | resolved |

## S020 (2013) — Students selfexplanations while solving unfamiliar cases: the role of biomedical

**CMOC** `S020-cmoc-f6acb6` · polarity: **negative** · verifier support: 0.42

> *When cases are already familiar heart failure cases, self-explanation diagnostic case solving still provides opportunities to generate explanations to oneself, but is associated with reduced engagement and elaboration relative to less familiar cases; the reported benefit was only demonstrated for unfamiliar cases.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | familiar heart failure cases | Heart failure was identified as a familiar topic and jaundice as a less familiar topic. | resolved |
| Intervention | self-explanation diagnostic case solving | They were then asked to read each case while self-explaining aloud for 8 minutes and then they had 2 minutes to write down their diagnosis, together with two su | resolved |
| Mechanism_Resource | opportunity to generate explanations to oneself | Self-explanation is a constructive learning strategy that consists of generating explanations to oneself in response to learning material. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Response | reduced engagement and elaboration in familiar cases | Students produced more self-explanation segments on less familiar cases than on familiar ones. | resolved |
| Outcome | benefit only demonstrated for unfamiliar cases | However, this positive effect was only demonstrated with unfamiliar clinical cases. | resolved |

## S021 (2013) — To what extent does nonanalytic reasoning contribute to visual learning in cytop

**CMOC** `S021-cmoc-45586b` · polarity: **positive** · verifier support: 0.84

> *When cytology novice participants receive annotated images, feature-based diagnostic support, and analytic feature-analysis instruction, these resources trigger feature-focused diagnostic analysis and greater readiness to report abnormalities, leading to improved discrimination and reduced miss errors.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | cytology novice participants | Forty-nine cytology novice participants undertook an initial image interpretation test to obtain baseline diagnostic accuracy (test 1). | resolved |
| Intervention | analytic cytomorphology training | Twenty-four participants subsequently received training in basic cervical cytomorphology and were given a list of cell features for future reference (the “analy | resolved |
| Mechanism_Resource | analytic feature-analysis instruction | Before test 2, participants were instructed to base their diagnosis solely on an analysis of the cytological features observed. | resolved |
| Mechanism_Resource | annotated normal/abnormal image pairs | Each image pair comprised 1 normal and 1 abnormal cell, and images were annotated with the relevant diagnostic features (Fig. 2). | resolved |
| Mechanism_Resource | feature-based diagnostic support | Participants in this group were also given a typed list of features for future reference, and a full explanation of the relevant cytological terminology was pro | resolved |
| Mechanism_Response | feature-focused diagnostic analysis | participants were instructed to base their diagnosis solely on an analysis of the cytological features observed. | resolved |
| Mechanism_Response | greater readiness to report abnormalities | When participants become aware of the features of cancer cells, discrimination should improve alongside a greater readiness to report cellular abnormalities. | resolved |
| Outcome | improved normal/abnormal discrimination | The increase in d 0 and decrease in criterion between tests 1 and 2 are consistent with improvements in the ability of participants to discriminate between norm | resolved |
| Outcome | reduced miss errors | Under both conditions, miss errors declined dramatically to 16.7% and 22.7%, respectively, following the respective training protocols (test 2). | resolved |

## S021 (2013) — To what extent does nonanalytic reasoning contribute to visual learning in cytop

**CMOC** `S021-cmoc-493325` · polarity: **positive** · verifier support: 0.78

> *When cytology novice participants receive non-feature-driven labelled exposure and are prompted to identify helpful features, they self-identify abnormal-cell features, leading to spontaneous diagnostic criteria generation with lay descriptions resembling standard cytopathology features.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | cytology novice participants | Forty-nine cytology novice participants undertook an initial image interpretation test to obtain baseline diagnostic accuracy (test 1). | resolved |
| Intervention | nonanalytic paired-image training | The remaining 25 participants were simply shown 20 nonannotated paired images of normal and abnormal cervical cells (the “nonanalytic” group). | resolved |
| Mechanism_Resource | non-feature-driven labelled exposure | no feature-driven training was provided, and the only image annotation used during training was restricted to the labels “normal” and “abnormal” | resolved |
| Mechanism_Resource | prompt to identify helpful features | Following test 2, participants were asked to write a list of features they felt were helpful in making their diagnostic decisions. | resolved |
| Mechanism_Response | self-identified helpful abnormal-cell features | features that participants in the nonanalytic group claimed were helpful in identifying abnormal cells, together with the frequency with which they were quoted. | resolved |
| Outcome | lay descriptions resembled standard features | Although written in layman’s terms, the morphologic descriptions are surprisingly similar to the standard list of features provided by the trainer to the analyt | resolved |
| Outcome | spontaneous diagnostic criteria generation | Our study clearly illustrates the extent to which inexperienced observers are able to generate diagnostic criteria without assistance from an expert trainer and | resolved |

## S021 (2013) — To what extent does nonanalytic reasoning contribute to visual learning in cytop

**CMOC** `S021-cmoc-a22fb7` · polarity: **negative** · verifier support: 0.62

> *When cytology novice participants are given a combined analytic/nonanalytic reasoning prompt whose meaning is not entirely clear, the instruction may trigger confusion, leading to decreased diagnostic accuracy and a significant decline in d prime.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | cytology novice participants | Forty-nine cytology novice participants undertook an initial image interpretation test to obtain baseline diagnostic accuracy (test 1). | resolved |
| Intervention | combined reasoning diagnostic instruction | Prior to a final test (test 3), participants in both groups were instructed to adopt a combined analytic/nonanalytic diagnostic strategy. | resolved |
| Mechanism_Resource | combined analytic and non-analytic reasoning prompt | participants in both groups were instructed to adopt a combined analytic/nonanalytic diagnostic strategy. | resolved |
| Mechanism_Resource | ambiguous combine-strategies instruction | We readily acknowledge that the meaning and validity of an instruction to “combine” analytic and nonanalytic reasoning is not entirely clear | resolved |
| Mechanism_Response | possible confusion about combined instruction | An alternative explanation for the relatively poor performance in the final test, which occurred in both groups of participants, is the possibility that partici | resolved |
| Outcome | significant decline in d prime | participants in both training groups showed a significant decline in d 0 in the final test. | resolved |
| Outcome | decreased diagnostic accuracy after combined instruction | Diagnostic accuracy in both groups improved significantly between tests 1 and 2 ( P<.001) but decreased between tests 2 and 3 ( P<.05). | resolved |

## S021 (2013) — To what extent does nonanalytic reasoning contribute to visual learning in cytop

**CMOC** `S021-cmoc-a278d2` · polarity: **positive** · verifier support: 0.42

> *When cytology novice participants receive nonannotated normal/abnormal image pairs and a first-impression similarity prompt, these resources trigger early nonanalytic reasoning, leading to intended nonanalytic pattern-recognition skill development, improved diagnostic performance, and more efficient responses.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | cytology novice participants | Forty-nine cytology novice participants undertook an initial image interpretation test to obtain baseline diagnostic accuracy (test 1). | resolved |
| Intervention | nonanalytic paired-image training | The remaining 25 participants were simply shown 20 nonannotated paired images of normal and abnormal cervical cells (the “nonanalytic” group). | resolved |
| Mechanism_Resource | first-impression similarity prompt | participants undertook test 2, for which they were asked to provide their first impression of each image based on their similarity to images seen during trainin | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Resource | nonannotated normal/abnormal image pairs | The remaining 25 participants were simply shown 20 nonannotated paired images of normal and abnormal cervical cells (the “nonanalytic” group). | resolved |
| Mechanism_Response | early nonanalytic reasoning | a significant element of nonanalytic reasoning from the earliest learning stages. | resolved |
| Outcome | improved diagnostic performance after brief exposure | After only brief exposure to paired images of normal and abnormal cells in the style of alternative forced choice training, the diagnostic performance of partic | resolved |
| Outcome | more efficient nonanalytic responses | response time data suggest that nonanalytic strategies are more efficient than rule-based training. | resolved |
| Outcome | nonanalytic pattern-recognition skill development | The intention, therefore, was for participants to develop nonanalytic pattern recognition skills without assistance from the trainer. | resolved |

## S021 (2013) — To what extent does nonanalytic reasoning contribute to visual learning in cytop

**CMOC** `S021-cmoc-d91e9d` · polarity: **negative** · verifier support: 0.78

> *When cytology novice participants receive exhaustive feature-list reasoning and feature-based diagnostic support, this triggers careful systematic feature consideration, which is associated with no diagnostic advantage over nonanalytic training and lower efficiency.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | cytology novice participants | Forty-nine cytology novice participants undertook an initial image interpretation test to obtain baseline diagnostic accuracy (test 1). | resolved |
| Intervention | analytic cytomorphology training | Twenty-four participants subsequently received training in basic cervical cytomorphology and were given a list of cell features for future reference (the “analy | resolved |
| Mechanism_Resource | feature-based diagnostic support | Participants in this group were also given a typed list of features for future reference, and a full explanation of the relevant cytological terminology was pro | resolved |
| Mechanism_Resource | exhaustive feature-list reasoning | cytopathology training usually involves the exhaustive application of morphological feature lists and diagnostic “rules,” all of which must be carefully and sys | resolved |
| Mechanism_Response | careful systematic feature consideration | all of which must be carefully and systematically considered before arriving at a diagnosis. | resolved |
| Outcome | lower efficiency than nonanalytic training | response time data suggest that nonanalytic strategies are more efficient than rule-based training. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Outcome | longer analytic response times after training | In the analytic training group, the median time for participants to respond “normal” or “abnormal” to each image increased from 2.53 seconds to 2.89 seconds bet | resolved |
| Outcome | no diagnostic advantage over nonanalytic training | Overall, the diagnostic performance of participants was no better following rule-based training than after a brief period of nonanalytic pattern recognition tra | UNRESOLVED |

## S022 (2014) — A Serious Game Can Be a Valid Method to Train Clinical

**CMOC** `S022-cmoc-197d52` · polarity: **negative** · verifier support: 0.78

> *When learners are novice medical students with little clinical experience, fixed timed case challenges with feedback can trigger frustration, leading to the level of difficulty being too high for some students.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | novice medical students with little clinical experience | interns (has experience as physician at surgical wards, no surgical training), master students (medical student in surgical clerkship), and bachelor students (m | resolved |
| Intervention | Medialis serious game | The serious game Medialis (Little Chicken Game Co., Amsterdam, The Netherlands) contains cases on diagnosis and management of patients with biliary tract diseas | resolved |
| Mechanism_Resource | fixed timed case challenge with feedback | The player had a maximum of 10 s to solve each case. After each attempt, players received feedback focused on their solution. | resolved |
| Mechanism_Response | novice frustration | Of the students, 8 of 17 found the game frustrating (compared to 1 of 11 trainees and 1 of 6 experts). | resolved |
| Outcome | difficulty too high for some students | According to the ‘‘flow theory,’’ the level of difficulty is too high for some students, which could be adjusted by incorporating difficulty levels. | resolved |

## S022 (2014) — A Serious Game Can Be a Valid Method to Train Clinical

**CMOC** `S022-cmoc-6b8431` · polarity: **positive** · verifier support: 0.42

> *When many surgical clinicians already have video-game experience, serious-game design for voluntary engagement triggers enjoyment, challenge, and involvement, leading to willingness to recommend the game and play in free time.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | video-game-experienced surgical clinicians | The majority of surgical residents and surgeons have experience with video games and are therefore likely to embrace using a video game in surgical training. | resolved |
| Intervention | serious game for surgical training | A serious game was developed to train surgical residents in clinical decision-making regarding biliary tract disease. | resolved |
| Mechanism_Resource | game design for voluntary engagement | A well-designed serious game combines psychological factors, design, and technology to engage learners in voluntary training. | resolved |
| Mechanism_Response | enjoyment and challenge | The majority of gamers had enjoyed the game, considering it fun (91.2 %) and challenging (85.3 %). | resolved |
| Mechanism_Response | felt involvement during gameplay | Furthermore, 44.1 % said they felt involved during game play (Supplementary Table 3). | resolved |
| Outcome | willingness to recommend and play voluntarily | The majority (81.8 %) would recommend it to colleagues, and 58.8 % would play in their free time. | resolved |

## S022 (2014) — A Serious Game Can Be a Valid Method to Train Clinical

**CMOC** `S022-cmoc-78e560` · polarity: **negative** · verifier support: 0.42

> *When clinically relevant expertise groups are compared for assessment, timer-based case solving is designed to trigger intrinsic motivation under time pressure, but the measured time required to solve cases is not a valid assessment parameter.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | expertise groups compared for assessment | Performances of user groups with clinically relevant levels of expertise were compared (surgeons, surgical residents, interns, master-degree students, bachelor- | UNRESOLVED |
| Intervention | serious game for decision assessment | A serious game was developed to train correct diagnosis and management of biliary tract disease. | resolved |
| Mechanism_Resource | timer-based case solving | The player had a maximum of 10 s to solve each case. | resolved |
| Mechanism_Response | intended intrinsic motivation under time pressure | This serious game is based on a quiz game in which time-pressure and competition is used to trigger the intrinsic motivation of the player [ 11]. | resolved |
| Outcome | decision speed not valid assessment parameter | Quality of decision-making appears to be a valid assessment parameter in the game (proportion correct), whereas the time required to solve cases is not. | resolved |

## S022 (2014) — A Serious Game Can Be a Valid Method to Train Clinical

**CMOC** `S022-cmoc-943a18` · polarity: **positive** · verifier support: 0.42

> *When everyday clinical practice is constrained by patient-safety concerns, repeated serious-game practice supports repeated play and memory storage, leading to improved performance in a second gaming session.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | patient-safety constraints on workplace practice | Patent safety concerns limit possibilities to train in everyday clinical practice. Therefore, educators seek alternative methods [ 2, 3]. | resolved |
| Intervention | Medialis serious game | The serious game Medialis (Little Chicken Game Co., Amsterdam, The Netherlands) contains cases on diagnosis and management of patients with biliary tract diseas | resolved |
| Mechanism_Resource | repeated practice to mastery | They assist in learning at different paces and allow practice to the point of mastery and automaticity. Repetition is a precondition for long-term potentiation. | resolved |
| Mechanism_Response | memory storage through repeated play | Whereas gamers repeat actions as they play, their strengthening of synaptic connections induces memory storage and learning. | resolved |
| Outcome | improved second-session performance | Figure 4 shows that participants improved their performance during their second gaming session (median proportion of 0.72 (IQR 0.23) compared to 0.48 (IQR 0.22) | resolved |

## S022 (2014) — A Serious Game Can Be a Valid Method to Train Clinical

**CMOC** `S022-cmoc-c871b1` · polarity: **positive** · verifier support: 0.62

> *When participants have different levels of biliary-disease expertise, realistic biliary decision cases trigger use of clinical experience in case solving, leading to decision accuracy that discriminates expertise levels.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | different expertise levels in biliary disease | Scores were compared between groups of different levels of expertise: surgeons (licensed surgeon), residents (surgical trainee), interns (has experience as phys | resolved |
| Intervention | Medialis serious game | The serious game Medialis (Little Chicken Game Co., Amsterdam, The Netherlands) contains cases on diagnosis and management of patients with biliary tract diseas | resolved |
| Mechanism_Resource | practice with realistic biliary decision cases | Cases consisted of an image, information describing a clinical problem, and possible solutions (Fig. 1). | resolved |
| Mechanism_Response | clinical experience applied to case solving | In total, 60.6 % stated that decisions in the serious game are based on realistic elements, and 66.7 % found their clinical experience to be helpful. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Outcome | expertise-discriminating decision accuracy | Surgeons solved more cases correctly (mean 77 %) than surgical residents (67 %), interns (60 %), master-degree students (50 %), and bachelor-degree students (39 | UNRESOLVED |

## S023 (2014) — Casebook: a virtual patient iPad application for teaching decision-making throug

**CMOC** `S023-cmoc-13194d` · polarity: **positive** · verifier support: 0.86

> *When undergraduate medical students are learning diagnostic reasoning, Casebook provides real patient EHR cases with explicit modelling of physician reasoning, triggering understanding of physician thought processes, leading to intended learning of decision-making skills.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | undergraduate medical students learning diagnostic reasoning | The foremost aim of the Casebook Virtual Patient application is to help to teach clinical diagnostic reasoning and decision-making skills to undergraduate stude | resolved |
| Intervention | Casebook virtual patient application | Casebook is a tablet-based, interactive Virtual Patient teaching application for undergraduate students of medicine. | resolved |
| Mechanism_Resource | practice with real patient EHR cases | Because the cases consist of data and patient records from real-life medical situations that genuinely occurred, the cases should more closely mimic real-life s | resolved |
| Mechanism_Resource | explicit modelling of physician reasoning | Each health record can also be annotated with meta-information by the case designer, such as insight into the thought processes and the decision-making rational | resolved |
| Mechanism_Response | understanding physician thought processes | we hope that students can gain an understanding into the thought processes and clinical reasoning of the physician who worked on the patient. | resolved |
| Outcome | decision-making skills learned | Students learn decision-making skills by observing and interacting with real patient cases in this simulated environment. | resolved |

## S023 (2014) — Casebook: a virtual patient iPad application for teaching decision-making throug

**CMOC** `S023-cmoc-ca7120` · polarity: **positive** · verifier support: 0.78

> *When diagnostic errors have been discovered in records, viewing diagnostic-error cases provides annotated records of mistakes and consequences, triggering students to examine error causes and sequences, leading to an intended reduction in diagnostic error.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | discovered diagnostic errors in records | Diagnostic error, discovered after the end of the patient 's treatment, would make for suitable cases to teach the consequences of diagnostic error | resolved |
| Intervention | viewing diagnostic-error cases | we recommend that cases that are known to have involved diagnostic error should also be viewed by students | resolved |
| Mechanism_Resource | annotated records of mistakes and consequences | health records, properly organised into patient cases, with the relevant annotations and discussions, provide for a unique opportunity to examine mistakes and t | resolved |
| Mechanism_Response | examining error causes and sequences | This would allow students to examine the cause of an error, or perhaps more importantly, the course of events that occurred which eventually led to an error. | resolved |
| Outcome | intended reduced diagnostic error | students can reduce diagnostic error when they have been provided the opportunity to learn from their mistakes | resolved |

## S023 (2014) — Casebook: a virtual patient iPad application for teaching decision-making throug

**CMOC** `S023-cmoc-d1780a` · polarity: **positive** · verifier support: 0.84

> *When students have limited access to common patient presentations, Casebook's EHR-based case-pool creation provides access to multitudes and collections of similar cases, triggering pattern recognition and illness-script development, leading to intended suitable and valid diagnoses.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | limited access to common patient presentations | students have less access to patients exhibiting common conditions, and it is these types of encounters we wish to simulate using the Casebook application | resolved |
| Intervention | Casebook EHR-based case-pool creation | The Casebook application aids the case creator by allowing for EHRs to be used as the basis for Virtual Patient cases. | resolved |
| Mechanism_Resource | collections of similar cases | pools of similar cases are required so that students can properly develop their illness scripts in order to make suitable and valid diagnoses when confronted wi | resolved |
| Mechanism_Resource | access to multitudes of patient cases | Learning theory research has shown that students learn decision-making skills best when they have access to multitudes of patient cases and it is this plurality | resolved |
| Mechanism_Response | pattern recognition from repeated cases | This experience builds pattern recognition, meaning the more experience a student has, the more likely they will make a correct diagnosis from the data availabl | resolved |
| Mechanism_Response | illness-script development | The student must have experience with similar patients or scenarios in order to build illness scripts: the combination of conditions, symptoms, and diseases tha | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Outcome | intended suitable and valid diagnoses | students can properly develop their illness scripts in order to make suitable and valid diagnoses when confronted with a new patient. | resolved |

## S023 (2014) — Casebook: a virtual patient iPad application for teaching decision-making throug

**CMOC** `S023-cmoc-e74ea7` · polarity: **positive** · verifier support: 0.78

> *When undergraduate students are learning clinical reasoning, Casebook's progress measurement during case navigation provides decision-point questions, immediate feedback, and the real physician decision, triggering a grasp of the possible diagnostic path, leading to diagnostic reasoning being tested rather than factual knowledge alone.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | undergraduate clinical reasoning learners | The foremost aim of the Casebook Virtual Patient application is to help to teach clinical diagnostic reasoning and decision-making skills to undergraduate stude | resolved |
| Intervention | Casebook progress measurement during navigation | Third, the software must examine the student as they navigate through the case, so that their progress can be measured and compared. | resolved |
| Mechanism_Resource | immediate feedback on answers | Once the user has submitted their answer, they are provided with immediate feedback and are allowed to continue to the next patient record | resolved |
| Mechanism_Resource | real physician decision revealed | When a student answers a question, they are presented with the patient record that shows what decision the physician made in the real world | resolved |
| Mechanism_Resource | decision-point questions | at strategic points throughout the case, questions must be answered by the student before they can continue. | resolved |
| Mechanism_Response | grasp of possible diagnostic path | whether the student has a grasp of what may be the correct path to take or diagnosis to make at a particular point of the patient 's treatment. | resolved |
| Outcome | diagnostic reasoning tested | cases should be used to test the diagnostic reasoning of the student, not to test their knowledge of a particular subject. | resolved |

## S024 (2014) — Ineffectiveness of cognitive forcing strategies

**CMOC** `S024-cmoc-a70983` · polarity: **negative** · verifier support: 0.62

> *When availability-bias cases contained a possible common but incorrect diagnosis and an uncommon correct diagnosis, instruction provided a warning about availability bias, yet learners still identified common incorrect diagnoses, leading to no improvement in uncommon diagnosis identification.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | availability-bias cases with uncommon correct diagnoses | Availability bias was tested with one case using an x-ray and one case using an ECG, for which a common diagnosis was possible but incorrect, whereas an uncommo | resolved |
| Intervention | availability-bias cognitive forcing instruction | Students were taught two biases: search satisficing and availability. With SSB, the clinician may identify one diagnosis but miss a second. | resolved |
| Mechanism_Resource | warning about common-diagnosis availability bias | With availability bias, clinicians may identify incorrect, common diagnoses and miss a correct, but uncommon, diagnosis. | resolved |
| Mechanism_Response | common incorrect diagnosis identification | Group Identified common incorrect diagnosis (%) Identified uncommon correct diagnosis (%) CFS training 141 (97.2) 66 (45.2) Control 41 (89.1) 21 (44.6) | resolved |
| Outcome | no improvement in uncommon diagnosis identification | In the availability bias cases, only 45% of the participants in both groups identified the uncommon correct diagnosis, with no difference between groups | resolved |

## S024 (2014) — Ineffectiveness of cognitive forcing strategies

**CMOC** `S024-cmoc-b2900b` · polarity: **negative** · verifier support: 0.84

> *When novices may lack sufficient experience to employ nonanalytic reasoning, CFS training provides metacognitive bias-identification techniques, but their reliance on analytic feature-counting constrains benefit, leading to little benefit from CFS.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | novices lacking nonanalytic reasoning experience | The overall diagnostic accuracy of both experimental and control groups suggests that these ‘‘novices’’ may lack sufficient experience to employ nonanalytic rea | resolved |
| Intervention | CFS training for diagnostic error reduction | The objective of this study was to determine the effect of CFS training on diagnostic error in senior medical students. | resolved |
| Mechanism_Resource | metacognitive bias-identification techniques | Specifically, these programs claimed that metacognitive techniques (termed cognitive forcing strategies [CFS]) can identify specific biases, promote self-monito | resolved |
| Mechanism_Response | reliance on analytic feature-counting | There is some evidence that novices do tend to rely on conscious, analytic reasoning, employing algorithms or feature counts to determine a diagnosis. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Outcome | little benefit from CFS | If novices are not using nonanalytic reasoning, CFS may be of little benefit. | resolved |

## S024 (2014) — Ineffectiveness of cognitive forcing strategies

**CMOC** `S024-cmoc-bc7021` · polarity: **negative** · verifier support: 0.78

> *When novice senior medical students received a cognitive forcing strategies seminar, the seminar provided self-monitoring and bias-countering strategy, but triggered no greater second-diagnosis search than control, leading to no significant improvement in second-diagnosis accuracy.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | novice senior medical students | The objective of this study was to determine the effect of CFS training on diagnostic error in senior medical students. | resolved |
| Intervention | cognitive forcing strategies seminar | The educational intervention consisted of a 90-minute, interactive, case-based seminar of 15 students and one instructor following an outline of standardized Mi | resolved |
| Mechanism_Resource | self-monitoring and bias-countering strategy | CFS teaching included self-monitoring during decision making, potential bias identification, and active employment of a cognitive process to counter a potential | resolved |
| Mechanism_Response | no greater second-diagnosis search | For the SSB cases, 52% of students with CFS training and 48% in the control group initiated a search for the second diagnosis | resolved |
| Outcome | no significant improvement in second-diagnosis accuracy | Of these, 54% correctly identified the second diagnosis in the CFS group and 48% did so in the control group; the difference was not significant | resolved |

## S024 (2014) — Ineffectiveness of cognitive forcing strategies

**CMOC** `S024-cmoc-c87221` · polarity: **negative** · verifier support: 0.28

> *When false-positive cases had no correct second diagnosis, search-satisficing CFS training provided SSB missed-second-diagnosis teaching; oversearching occurred in both groups, leading to incorrect second diagnoses without CFS excess and no significant oversearching difference.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | false-positive cases with no second diagnosis | Two ‘‘false positive’’ search satisficing cases, for which there is no correct second diagnosis, were also included to determine if students were ‘‘oversearchin | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Intervention | search-satisficing CFS training | Application of CFS could enable reduction of diagnostic error, as evidenced by identifying multiple correct diagnoses for the two cases prone to search satisfic | resolved |
| Mechanism_Resource | SSB missed-second-diagnosis teaching | With SSB, the clinician may identify one diagnosis but miss a second. | resolved |
| Mechanism_Response | second-diagnosis oversearching in both groups | False positive cases were designed to see if CFS training promoted oversearching for the second diagnosis. | resolved |
| Outcome | no significant oversearching difference | There were no significant differences between groups (x2 5 2.38, df 5 1, p 5 0.12; Table 3). | resolved |
| Outcome | incorrect second diagnoses without CFS excess | Sixty-four percent of the CFS group and 77% of the control group incorrectly identified a second diagnosis. | resolved |

## S024 (2014) — Ineffectiveness of cognitive forcing strategies

**CMOC** `S024-cmoc-f19021` · polarity: **negative** · verifier support: 0.62

> *When analytic processing is limited by working memory and cognitive load, analytic CFS training provides an active cognitive process to counter bias, which the authors hypothesise may overwhelm cognitive load, leading to no impact on diagnostic error prevention.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | limited working memory and cognitive load | Most analytic processes are limited by the amount of working memory of the learner, as well as the intrinsic and extrinsic cognitive load of the activity. | resolved |
| Intervention | analytic CFS training | CFS teaching included self-monitoring during decision making, potential bias identification, and active employment of a cognitive process to counter a potential | resolved |
| Mechanism_Resource | active cognitive process to counter bias | active employment of a cognitive process to counter a potential bias. | resolved |
| Mechanism_Response | hypothesised overwhelmed cognitive load | may simply overwhelm an individual's cognitive load without any effect on outcomes. | resolved |
| Outcome | no impact on diagnostic error prevention | CFS training (as implemented in this study) had no impact on preventing diagnostic error. | resolved |

## S025 (2015) — Does medical students' diagnostic performance improve by observing examples of s

**CMOC** `S025-cmoc-420a84` · polarity: **negative** · verifier support: 0.62

> *When early clerkship students observe expert self-explanations, expert reasoning with skipped and abstract steps may trigger difficulty relating the reasoning to their own knowledge, leading to expert models being no more effective than peer models.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | early clerkship medical students | Students in their first 8 months of clerkship, in two consecutive cohorts (academic year 2011–12, and academic year 2012–13), were invited | resolved |
| Intervention | listening to expert SE examples | They solved a set of 4 clinical cases twice: after having self-explained and then again after having listened to an example of SE from a peer or an expert | resolved |
| Mechanism_Resource | expert reasoning with skipped abstract steps | Experts use fast, largely non-analytic reasoning, tend to skip steps and to reason at a higher level of abstraction | resolved |
| Mechanism_Response | difficulty relating expert reasoning to own knowledge | which may be difficult for students to relate to their own knowledge. | resolved |
| Outcome | expert models not more effective than peer models | expert SE models were not more effective than peer SE models | resolved |

## S025 (2015) — Does medical students' diagnostic performance improve by observing examples of s

**CMOC** `S025-cmoc-4bf1c7` · polarity: **negative** · verifier support: 0.42

> *When example observation lacks active-processing prompts, listening to natural raw self-explanation examples may trigger passive listening, leading to no added value on transfer cases.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | no active-processing prompts for examples | They were not given prompts or other specific tasks to ensure that they listened actively to the examples. | UNRESOLVED |
| Intervention | listening to self-explanation examples | Participants were informed that they would listen to an example of SE from someone else and that they would have to solve the problem again. | resolved |
| Mechanism_Resource | natural raw self-explanation examples | In contrast, in our study, we used natural ‘‘raw’’ examples of SE, as they might occur in clinical setting. | resolved |
| Mechanism_Response | passive listening to examples | They may have been passive, even though their prior SE on the same case was expected to act as a preparatory task | resolved |
| Outcome | no added value on transfer cases | For transfer cases 1 week later, no added value of SE models was noted. | resolved |

## S025 (2015) — Does medical students' diagnostic performance improve by observing examples of s

**CMOC** `S025-cmoc-a5bac6` · polarity: **positive** · verifier support: 0.78

> *When cases concern a less familiar clinical topic, self-explanation diagnostic case solving provides opportunities to generate explanations to oneself, triggering active thinking while self-explaining and leading to improved diagnostic accuracy and performance.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | less familiar jaundice cases | Jaundice was a rather unfamiliar topic to the 3rd-year medical students involved (Chamberland et al. 2011). | resolved |
| Intervention | self-explanation diagnostic case solving | they were asked to read each case, generate SE out loud for 8 min | resolved |
| Mechanism_Resource | opportunity to generate explanations to oneself | learners generate for themselves explanations about pieces of information in a text to be learned or a problem to be solved | resolved |
| Mechanism_Response | active thinking while self-explaining | lack of familiarity seemed to act as a trigger for medical students to engage in more active thinking while self-explaining for diagnosing cases | resolved |
| Outcome | improved diagnostic accuracy and performance | students' diagnostic accuracy and diagnostic performance improved significantly across the study phases relative to their initial performance | resolved |

## S025 (2015) — Does medical students' diagnostic performance improve by observing examples of s

**CMOC** `S025-cmoc-c34f6a` · polarity: **negative** · verifier support: 0.78

> *When third-year medical students observe same-level peers, peer self-explanation examples containing irrelevant details and uncorrected errors may trigger little learning from those examples, leading to no performance advantage by model type.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | third-year medical student observers | Participants were 53 third-year medical students from the Universite´ de Sherbrooke in Que´bec, Canada. | resolved |
| Intervention | listening to peer SE examples | Participants of the peer-SE and expert-SE groups were then instructed to listen to an 8-min recording of either peer or expert SE | resolved |
| Mechanism_Resource | peer examples with errors and irrelevant details | We used examples from peers (third-year medical students), whose SE included irrelevant details and even uncorrected errors. | resolved |
| Mechanism_Response | little learning from same-level examples | Despite their cognitive proximity, participants at the exact same level may have little to learn from such examples. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Outcome | no performance difference by model type | the type of SE model (peer vs. expert) did not make any difference on students' performance | resolved |

## S025 (2015) — Does medical students' diagnostic performance improve by observing examples of s

**CMOC** `S025-cmoc-ef96a0` · polarity: **positive** · verifier support: 0.72

> *When students have already self-explained the same cases, listening to peer or expert self-explanation examples provides immediate feedback on reasoning, triggering critical revision of knowledge representations and leading to an immediate maintained gain in diagnostic accuracy.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | prior self-explanation of same cases | The present study investigated the impact on students' diagnostic performance of listening to peer or expert SE examples after having themselves self-explained | resolved |
| Intervention | listening to peer or expert SE examples | Participants of the peer-SE and expert-SE groups were then instructed to listen to an 8-min recording of either peer or expert SE | resolved |
| Mechanism_Resource | immediate feedback on reasoning | SE examples presumably provide some form of immediate feedback to students, which may explain why they improved already in the training phase. | resolved |
| Mechanism_Response | critical revision of knowledge representations | observing explicit examples would give additional opportunities for students to extend their understanding and critically revise their own knowledge representat | resolved |
| Outcome | immediate maintained diagnostic accuracy gain | Combining SE models with students' own SE led to an immediate improvement in diagnostic accuracy that was maintained after 1 week | resolved |

## S026 (2015) — Self-explanation in learning clinical reasoning: the added value of examp

**CMOC** `S026-cmoc-15e102` · polarity: **positive** · verifier support: 0.78

> *When Year 3 clerks work with less familiar jaundice cases, self-explanation diagnostic case solving provides opportunities to generate explanations to oneself, triggering knowledge elaboration and biomedical-clinical linking, leading to near-transfer diagnostic gains.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | less familiar jaundice cases | All cases were related to jaundice because this clinical topic was rather unfamiliar to the Year 3 medical students involved. | resolved |
| Intervention | self-explanation diagnostic case solving | In each group, the students first solved four clinical cases using SE. Then, Group 1 listened to examples of residents' SEs for the same cases | resolved |
| Mechanism_Resource | opportunity to generate explanations to oneself | Self-explanation is an active learning process that consists of generating explanations to oneself when working through learning materials. | resolved |
| Mechanism_Response | reactivating biomedical and clinical knowledge links | Unfamiliar cases seem to encourage students to reactivate biomedical knowledge, potentially creating new links between biomedical and clinical knowledge | resolved |
| Mechanism_Response | knowledge elaboration integration and monitoring | While self-explaining, learners engage in knowledge elaboration, integration and monitoring, allowing them to revise their knowledge and build a more coherent r | resolved |
| Outcome | near-transfer diagnostic gains | all groups, including the control group (SE alone), improved significantly in both diagnostic accuracy and diagnostic performance scores on near-transfer cases. | resolved |

## S026 (2015) — Self-explanation in learning clinical reasoning: the added value of examp

**CMOC** `S026-cmoc-20c24b` · polarity: **positive** · verifier support: 0.78

> *When resident reasoning is near-peer and cognitively congruent, resident SE examples with prompts provide relatable modelling and prompts, triggering comparison plus additional self-explanation and deep processing, leading to far-transfer diagnostic performance gains.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | near-peer cognitive congruence | Junior residents represent near peers for medical clerks. Their clinical reasoning is close to, or cognitively congruent with, the students' own | resolved |
| Intervention | resident SE examples with prompts | Group 1 listened to examples of residents' SEs with prompts; Group 2 listened to examples of residents' SEs without prompts | resolved |
| Mechanism_Resource | justification and mental-model prompts | A total of 12 prompts, six of each type, were incorporated into the four resident SE recordings. | resolved |
| Mechanism_Resource | near-peer resident reasoning examples | At the same time, the residents' more developed reasoning acts as a source of new information from which students can learn. | resolved |
| Mechanism_Response | additional self-explanation and deep processing | Justification and mental model revision prompts presumably engaged students in additional SE while they listened to the example, thereby supporting the deep proc | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Response | relating to and comparing reasoning | Their clinical reasoning is close to, or cognitively congruent with, the students' own, which allows the latter to relate to it easily and to appreciate similar | resolved |
| Outcome | far-transfer diagnostic performance gains | Indeed, Group 1, with the combined intervention, performed significantly better on both scores compared with the control group. | resolved |

## S026 (2015) — Self-explanation in learning clinical reasoning: the added value of examp

**CMOC** `S026-cmoc-265835` · polarity: **mixed** · verifier support: 0.42

> *When near-transfer cases are similar to the training cases, the SE-alone control condition provides case self-explanation, which the authors suggest triggers deeper processing over the week, leading to delayed rather than immediate diagnostic improvement.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | near-transfer cases similar to training cases | The near-transfer cases were similar to the training cases and used the same key elements and final diagnoses. | resolved |
| Intervention | SE-alone control condition | the control group with SE alone and in the two other groups, who were exposed to example SEs, differed. | resolved |
| Mechanism_Resource | case self-explanation task | Then, the participant was asked to solve the first training case, while self-explaining aloud for 8 minutes. | resolved |
| Mechanism_Response | suggested deeper data processing over time | which again suggests deeper data processing in the course of a week. | resolved |
| Outcome | delayed diagnostic improvement | For students in the control group, no improvement was noted immediately, but a significant increase in scores occurred 1 week later. | resolved |

## S026 (2015) — Self-explanation in learning clinical reasoning: the added value of examp

**CMOC** `S026-cmoc-3f259e` · polarity: **positive** · verifier support: 0.84

> *When junior residents are cognitively congruent near peers, resident SE examples provide more developed but relatable reasoning, triggering students to elaborate, confirm and complete their diagnostic arguments, leading to accelerated and maintained improvement.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | near-peer cognitive congruence | Junior residents represent near peers for medical clerks. Their clinical reasoning is close to, or cognitively congruent with, the students' own | resolved |
| Intervention | resident SE examples | listening to the example of a resident's SE and being prompted while doing so improved students' diagnostic performance score on near-transfer cases | resolved |
| Mechanism_Resource | near-peer resident reasoning examples | At the same time, the residents' more developed reasoning acts as a source of new information from which students can learn. | resolved |
| Mechanism_Response | elaboration and reasoning confirmation | Students reported that listening to residents' examples helped them to elaborate further on their own ideas. | resolved |
| Mechanism_Response | completing diagnostic arguments with new ideas | They also reported that it helped them to confirm their reasoning and to complete their final diagnostic opinion and argument with some ideas | resolved |
| Outcome | accelerated maintained improvement | For the latter two groups, the improvement was immediate and maintained over time, whereas the group subjected to SE alone took 1 week | resolved |

## S026 (2015) — Self-explanation in learning clinical reasoning: the added value of examp

**CMOC** `S026-cmoc-48d0ac` · polarity: **negative** · verifier support: 0.42

> *When far-transfer cases are more difficult, resident SE examples without prompts provide the same recordings but lack prompt-supported additional self-explanation; because active processing appears critical, this pathway leads to no far-transfer performance advantage over control.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | more difficult far-transfer cases | The latter were also made more difficult by a less typical presentation, less common final diagnosis or the presence of comorbidity. | resolved |
| Intervention | resident SE examples without prompts | Participants in Group 2 listened to the 8-minute example of resident SE without prompts. They were advised that a few silent sequences were included | resolved |
| Mechanism_Resource | unprompted resident SE recordings | All participants in Groups 1 and 2 were exposed to the same recordings of residents' SEs. | resolved |
| Mechanism_Response | active processing critical for example learning | Students' exposure to adapted examples of clinical reasoning is important, but their active processing of these examples seems critical. | resolved |
| Mechanism_Response | less prompt-supported additional self-explanation | Justification and mental model revision prompts presumably engaged students in additional SE while they listened to the example, thereby supporting the deep proc | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Outcome | no far-transfer performance advantage over control | there was also a significant difference between Groups 1 and 2 (p = 0.018), but no difference between Group 2 and the control group | resolved |

## S027 (2016) — Test-enhanced learning of clinical reasoning: a crossover randomised tria

**CMOC** `S027-cmoc-06fbb7` · polarity: **negative** · verifier support: 0.78

> *When examinations are formative and no incentive is provided, assessments without credit points trigger students not to revise, contributing to disappointing overall retention performance.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | formative no-incentive assessment context | All examinations were formative in nature (i.e. no incentive to achieve high scores was provided as this may have confounded results 11). | resolved |
| Intervention | formative entry exit retention examinations | the same cases were used in the entry and exit examinations, as well as the retention test | resolved |
| Mechanism_Resource | assessments without credit points | As these assessments did not generate credit points, students did not revise for them | resolved |
| Mechanism_Response | students did not revise | students did not revise for them, which avoided any potential confounding imposed by the impact of examination consequences on learning behaviour. | resolved |
| Outcome | disappointing overall retention performance | At first glance, the overall performance in the retention test, which resulted in a mean score of just over 50%, is disappointing. | resolved |

## S027 (2016) — Test-enhanced learning of clinical reasoning: a crossover randomised tria

**CMOC** `S027-cmoc-729257` · polarity: **positive** · verifier support: 0.74

> *When Year 4 clinical medical students encounter repeated case-based learning, key-feature question prompts with spaced immediate feedback trigger storage and retrieval of knowledge, leading to higher long-term clinical reasoning scores.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | Year 4 clinical medical students | Year 4 students enrolled in three consecutive teaching modules (cardiology and pulmonology, nephrology and rheumatology, haematology and oncology) in the winter | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Intervention | formative key-feature case testing | repeated case-based learning augmented with formative key feature questions is more effective than case-based learning alone in fostering clinical reasoning in | resolved |
| Mechanism_Resource | spaced immediate feedback | Testing was spaced across weeks and students were given immediate educational feedback on their answers. | resolved |
| Mechanism_Resource | key-feature question prompts | the latter also involved prompts to answer five key feature questions per case. These questions are referred to as ‘items’; all analyses were performed at the it | resolved |
| Mechanism_Response | storage and retrieval of knowledge | the observed score difference can be attributed to an effect of repeated testing on the storage and retrieval of knowledge rather than an effect of training in | resolved |
| Outcome | higher long-term clinical reasoning scores | Mean /C6SD scores for test items and study-only items were 56.0 /C625.8% and 48.8 /C624.7%, respectively (p < 0.001). | resolved |

## S027 (2016) — Test-enhanced learning of clinical reasoning: a crossover randomised tria

**CMOC** `S027-cmoc-72bcfc` · polarity: **mixed** · verifier support: 0.86

> *When e-seminars are self-paced, additional questions in test cases trigger more time spent on test items; this has only weak association with better outcomes, and the testing effect persists after controlling for time-on-task.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | self-paced e-seminar setting | Examinations and e-seminars were scheduled to last no longer than 45 minutes within which students were free to complete sessions at their own pace. | resolved |
| Intervention | test-enhanced case presentation | The intervention condition used the same content but augmented case presentation with a sequence of key feature questions. | resolved |
| Mechanism_Resource | additional questions in test cases | the only difference between them was the presence of five questions (‘items’) in each of the latter. | resolved |
| Mechanism_Response | more time spent on test items | students spent significantly more time on test items than on study-only items (21.47 /C65.27 minutes versus 13.30 /C65.24 minutes per seminar; p < 0.001). | resolved |
| Outcome | weak association but persistent adjusted effect | There was weak evidence of an association between extra time-on-task in test sessions and better learning outcomes on test items, but the effect of the interven | resolved |

## S027 (2016) — Test-enhanced learning of clinical reasoning: a crossover randomised tria

**CMOC** `S027-cmoc-8f9b93` · polarity: **positive** · verifier support: 0.78

> *When students have answered an item incorrectly on a prior test, item-level feedback and explanations of incorrect distractors trigger learning from feedback, leading some students to correct that item on a subsequent test.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | prior incorrect item answer | students who failed to provide a correct answer in one test | resolved |
| Intervention | computer-assisted key-feature e-seminars | Students in both groups attended weekly computer-assisted learning sessions (‘e-seminars’; maximum duration: 45 minutes) during which they were presented with f | resolved |
| Mechanism_Resource | item-level feedback access | Detailed feedback was created for each item; students were able to access this feedback by clicking on a specific button. | resolved |
| Mechanism_Resource | explanations of incorrect distractors | It also contained explanations of why some distractor options anticipated to be chosen by students were deemed ‘incorrect’. | resolved |
| Mechanism_Response | learning from feedback | thus suggesting a learning effect elicited by the feedback provided. | resolved |
| Outcome | some students subsequently corrected errors | 33.3% of students who failed to provide a correct answer in one test went on to submit a correct answer in a subsequent test of the same item | resolved |

## S028 (None) — 18 Reprinted from Australian Family Physician Vol. 41, No. 1/2 january/february

**CMOC** `S028-cmoc-0a511e` · polarity: **positive** · verifier support: 0.86

> *When novice learners have limited clinical experience, teaching clinical reasoning within a consultation skill framework provides a deconstructed consultation focused on the reasoning pathway, triggering understanding of expert thinking and reasoning, leading to enhanced student clinical reasoning ability.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | novice learners with limited clinical experience | novice learners, such as medical students, have limited clinical experience and therefore need to approach most consultations in a more analytical (‘hypothetico | resolved |
| Intervention | clinical reasoning taught within consultation skill framework | A proposed strategy for teaching clinical reasoning is therefore to teach it within a skill framework in the clinical consultation room. | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Resource | deconstructed consultation focused on reasoning pathway | A deconstructed consultation focusing on the clinical reasoning pathway is presented in Figure 1. | resolved |
| Mechanism_Response | understanding expert thinking and reasoning | this is to assist the student to understand what the clinician was thinking and how they were reasoning. | resolved |
| Outcome | enhanced student clinical reasoning ability | clinical reasoning should not be considered as something tacit that students simply ‘pick up as they go along’, and clinical teachers can greatly enhance studen | resolved |

## S028 (None) — 18 Reprinted from Australian Family Physician Vol. 41, No. 1/2 january/february

**CMOC** `S028-cmoc-1490e5` · polarity: **positive** · verifier support: 0.62

> *When clinicians’ reasoning is automated in practice, teaching clinical reasoning provides an opportunity to review reasoning pathways, triggering reflection on their own reasoning, leading to improved diagnostic efficiency, fewer assumptions and investigations, and improved patient satisfaction.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | automated clinician reasoning in practice | Automation of clinical reasoning allows clinicians to undertake consultations in a timely and streamlined fashion, but does so at the risk of missing critical i | resolved |
| Intervention | teaching clinical reasoning for clinician reflection | Teaching clinical reasoning encourages personal reflection and refinement of the clinician’s own clinical reasoning skills to improve patient outcomes. | resolved |
| Mechanism_Resource | opportunity to review reasoning pathways | teaching clinical reasoning also provides an excellent opportunity for clinicians to consider and review their own clinical reasoning pathways. | resolved |
| Mechanism_Response | reflection on own reasoning | the benefits of reflecting upon and improving one’s own reasoning may include: | resolved |
| Outcome | improved diagnostic efficiency and patient satisfaction | improving time to diagnosis | |
| • avoiding assumptions | | | |
| • reducing unnecessary investigation and the costs these incur | | | |
| • improving patient satisfaction and being bra | resolved | | |

## S028 (None) — 18 Reprinted from Australian Family Physician Vol. 41, No. 1/2 january/february

**CMOC** `S028-cmoc-cb3c24` · polarity: **positive** · verifier support: 0.78

> *When a student is familiar with the clinician reasoning model, supervised performance consultation provides interjected questions when reasoning is unclear, triggering articulation of reasoning during the work-up, leading to the student being kept on track during performance.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | student familiar with clinician reasoning model | the student should now be familiar with how the clinician reasons through a clinical consultation. | resolved |
| Intervention | performance consultation with supervised student reasoning | in the third consultation, the student follows the model outlined in the first and second consultations (Figure 1) | resolved |
| Mechanism_Resource | interjected questions on unclear reasoning | the clinical teacher can interject with questions such as ‘Why did you ask that?’ or ‘How does that information help you?’ | resolved |
| Mechanism_Response | articulation of reasoning during work-up | performing the history, physical examination and directing investigations, explaining to the clinical teacher their reasoning as they proceed. | resolved |
| Outcome | student kept on track during performance | the clinical teacher can interject with questions such as ‘Why did you ask that?’ or ‘How does that information help you?’ to keep the student on track. | resolved |

## S028 (None) — 18 Reprinted from Australian Family Physician Vol. 41, No. 1/2 january/february

**CMOC** `S028-cmoc-d81042` · polarity: **positive** · verifier support: 0.78

> *When students have diagnostic knowledge but find patient-information navigation challenging, a coached comprehension consultation provides prompts about reasoning relevance, triggering active tracking of demonstrated reasoning, leading students to determine primary and differential diagnoses.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | students challenged by patient-information navigation | medical students are armed with knowledge helpful to patient diagnosis. but they can find it challenging to know how to navigate the information during a patien | resolved |
| Intervention | comprehension-phase coached consultation | following the demonstration and deconstruction phase, the next consultation should again be performed by the teaching clinician, but with the student actively t | resolved |

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Mechanism_Resource | coach prompts about reasoning relevance | The clinician may pause after certain critical questions to ask the student ‘Why did I ask that?’ or after a patient response to ask ‘How does that affect the h | resolved |
| Mechanism_Response | active tracking of demonstrated reasoning | with the student actively tracking the consultation and outlining the clinical reasoning being demonstrated. | resolved |
| Outcome | student determines primary and differential diagnoses | allow the student to determine the primary hypothesis and differential diagnosis following the history taking | resolved |

## S028 (None) — 18 Reprinted from Australian Family Physician Vol. 41, No. 1/2 january/february

**CMOC** `S028-cmoc-dcd47e` · polarity: **negative** · verifier support: 0.78

> *When there is an experience gap between practising clinicians and learners, a tacit pick-up approach provides only automated reasoning on display, triggering difficulty grasping the reasoning, leading to a barrier to learning clinical reasoning.*

| ELEMENT | LABEL | VERBATIM QUOTE | SPAN |
|---|---|---|---|
| Context | experience gap between clinician and learner | Clinical reasoning therefore requires not only an accumulation of knowledge but also a level of experience, which is generally what sets apart a practising clin | resolved |
| Intervention | tacit pick-up approach to clinical reasoning | clinical reasoning should not be considered as something tacit that students simply ‘pick up as they go along’ | resolved |
| Mechanism_Resource | automated reasoning on display | There is also a degree of automation2–4 that occurs when clinicians consult | UNRESOLVED |
| Mechanism_Response | difficulty grasping automated reasoning | There is also a degree of automation2–4 that occurs when clinicians consult, which to the observing student is difficult to grasp and can be a barrier to learni | resolved |
| Outcome | barrier to learning clinical reasoning | There is also a degree of automation2–4 that occurs when clinicians consult, which to the observing student is difficult to grasp and can be a barrier to learni | resolved |